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RESEARCH ARTICLE

Evolution of Layer1/Layer2-Triggered Mobility in 5G-Advanced: From Release 18 Foundation to Release 19 Enhancements and Beyond

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ABSTRACT This work provides a comprehensive examination of Layer 1/Layer 2-triggered mobility (LTM) enhancements introduced in 3rd Generation Partnership Project (3GPP) Release (Rel.) 19, extending the foundational framework established in Rel. 18. The key advancement is the extension of LTM to inter-central unit (CU) scenarios, significantly broadening its applicability across diverse network deployments. This extension necessitated comprehensive standardization work addressing security key management, dual connectivity (DC) operations, and channel state information (CSI) resource coordination mechanisms. The measurement capabilities were improved by incorporating channel state information reference signal (CSI-RS)-based layer 1 (L1)-reference signal received power (RSRP) measurements and implementing event-triggered reporting to reduce signaling overhead. Additionally, conditional LTM enhances mobility robustness through autonomous user equipment (UE)-triggered cell switching, while early CSI acquisition procedures optimize post-handover throughput performance. The paper elaborates on the technical implementation of inter-CU LTM, covering network signaling architectures, security framework adaptations, and DC integration aspects. Furthermore, we explore Rel. 20 developments focusing on LTM integration with Carrier Aggregation and adaptive measurement configuration mechanisms, alongside initial considerations for 6G mobility evolution. To emphasize the importance of reducing the handover interruption time, our system-level simulation analysis quantifies the transport layer benefits of reduced interruption times, demonstrating that reducing the interruption duration in LTM to approximately 20 ms delivers significant transmission control protocol (TCP) throughput gains over conventional handover's 100 ms interruption, especially when TCP re-transmission timeout (RTO) timers are set at 100 ms or below.

INDEX TERMS Mobility, handover, LTM, 3GPP, R19, 6G, RACH, INTER-CU, CSI-RS, CSI acquisition, security, event-based reporting, conditional LTM.

I. INTRODUCTION

In our previous paper [1], we have explained the LTM procedure of Rel. 18. We have shared that LTM was a merger of beam management and mobility enhancements to use the best of both worlds. We have explained that LTM was for

intra-CU and many interruption time reduction enhancements were introduced, such as transmission configuration indicator (TCI) state activation, early timing advance (TA) acquisition, early decoding and processing, and random access channel (RACH)-less LTM execution. The features of L1 RSRP measurement and reporting, and subsequent mobility are introduced to improve mobility robustness. In this paper, we aim to analyze the 3GPP Rel. 19 enhancements made to

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Rel. 18 LTM baseline solution in a way that is not discussed in the specifications. In doing so, we present each enhancement along with its motivation, what problem it solves and we identify the open challenges and point out to future directions for research. Rel. 19 LTM enhancements can be categorized in four main areas:

- Extension to wider deployment scenarios
- Resource overhead reduction
- Improved mobility robustness.
- Cell edge throughput improvements

More specifically, under extension to wider deployment scenarios area, we will introduce inter-CU LTM and related security enhancements. Under the resource overhead reduction, we introduce the event-based reporting framework to reduce the L1-RSRP reporting overhead. For mobility robustness, we consider CSI-RS-based L1-RSRP measurements, especially for frequency range 2 (FR2) deployments and conditional LTM. Finally, to improve the cell edge throughput, the procedure of early CSI acquisition is introduced.

LTM will be further enhanced in Rel. 20 with a focus on cell-edge throughput improvements and resource overhead reduction. In Rel. 20, LTM with carrier aggregation (CA) objective aims to enable better throughput during cell change, while dynamic measurement configuration is expected to reduce the measurement and reporting overhead. At the end, with the 6-th generation (6G) study item in Rel. 20, we expect to learn from the mistakes of complete 5-th generation (5G) and 5G-advanced (5GA). The aim would be to introduce a mobility procedure that fulfills the interruption time, mobility robustness, and cell edge throughput required to make 6G a commercial success.

To that end, in addition to introducing the Rel. 19 enhancements in LTM, in this paper, we also investigate the effect of interruption time reduction on transport layer performance. In fact, the effect of reducing interruption time on reliability is already discussed in [2] for three different cellular deployment scenarios, namely the future railway mobile communications system (FRMCS) scenario [3], a highway scenario and an urban macro scenario. The conclusion reached was that the effect of mobility procedures become critical for meeting high reliability. However, [2] focuses on radio access network (RAN) side, and it doesn't provide an end-to-end perspective. From a higher layer perspective, there are some measurement campaigns [4], [5] carried out in high-speed train environments that analyze the effect of extreme mobility on TCP performance. Their findings indicate that conventional transport layer congestion control algorithms can fail to adapt to rapidly changing radio conditions. Furthermore, there are some transport layer protocols designed to be used in cellular networks [6], [7] that provide performance gains when compared to more standard TCP variants, such as TCP CUBIC [8]. In [9], the effect of handovers on TCP performance in a 5G mmWave deployment is discussed. However, their focus is not on

the handover interruption time duration, since their reported average measured interruption time is 27.0 ms. As can be seen, although there is some prior research trying to improve the transport layer performance over cellular networks, and some work discussing the effect of handovers on TCP performance, to the best of authors' knowledge, there is not a discussion showing the effect of different interruption times induced by different mobility procedures and how LTM stacks up against them with the improvements it brings, which is the goal of our investigations.

The paper is organized as follows: Section II describes the Rel. 19 enhancements in depth. Section II-A introduces expansion of LTM procedure to Inter-CU, where signalling aspects, handling of configuration change such as layer 2 (L2) reset, security update and CSI resource configuration are described in detail in relevant subsections. Section II-B starts with an introduction of security fundamentals and configuration, this is followed by the problems of using the same security mechanism for subsequent inter-CU LTM procedure and finally the solutions introduced in Rel. 19 are shared. Section II-C discusses the expansion of DC LTM to inter-CU scenario covering inter-CU master cell group (MCG) LTM and inter-CU secondary cell group (SCG) LTM in detail. Section II-D introduces the measurement enhancement covering the CSI-RS based measurements and event triggered reporting and configuration and the extension of LTM to work with layer 3 (L3) measurements. Section II-E covers the novel procedure introduced for minimizing throughput degradation during mobility called early CSI acquisition. Last but not least, the conditional operation for LTM procedure is introduced in Section II-F and this concludes Section II. Section II-F introduces and discusses the 3GPP Rel. 20 work item that has just started in February 2026. This section also comprises a short summary of initial insights on 6G mobility procedure. Section IV introduces system level simulation results to investigate the impact of reduced interruption times of LTM procedure on transport layer throughput using TCP and aims at pinpointing what level of interruption time reduction is relevant from a transport layer point of view. Section V concludes the paper.

II. RELEASE 19 TOPICS

3GPP Rel. 19 introduced several critical enhancements to LTM that address the practical deployment challenges identified in Rel. 18. These improvements focus on expanding the applicability of LTM beyond single CU deployments, optimizing measurement procedures for better resource efficiency, and enhancing the overall robustness of the mobility framework. The key enhancement areas include inter-CU LTM support, measurement procedure optimizations, and advanced CSI acquisition mechanisms [10]. Each of these enhancements contributes to making LTM a more practical and deployable solution for real-world 5G networks.

A. INTER-CU LTM

3GPP Rel. 18 LTM introduced several features on top of radio resource control (RRC)-based mobility. For example, medium access control (MAC)-based cell switch triggering, L1-RSRP measurements for neighboring cells, early downlink (DL) and uplink (UL) synchronization and subsequent mobility, described in detail in [1]. The focus in the Rel. 18 was to support intra-gNodeB (gNB) (intra-CU) scenarios. To expand the applicability of LTM to a wider range of mobility scenarios, 3GPP decided to introduce inter-CU LTM to be specified in Rel. 19.

1) HANDOVER-RELATED NETWORK SIGNALING

RRC-based baseline HO may be prepared either via the Xn interface or via the NG interface. Xn interface provides a direct link between two gNBs in RAN, while NG interface provides a link between RAN gNB and the Core access and mobility management function (AMF). Inter-CU LTM specified in the Rel. 19 is supported only over Xn interface. LTM is by default a subsequent mobility mechanism, which means that the mobility configuration at the UE and the UE context at candidate nodes are maintained after a cell switch, to be used for a subsequent cell switch from the new serving cell to another candidate. This implies that Xn connectivity is required between each possible pair of gNBs included in the LTM configuration. Nevertheless, it should be noted that in typical network deployments an interface is available between gNBs served by the same AMF. A high-level visualization of the inter-CU signaling procedure is provided in Fig. 1. As can be seen in Fig. 1, a UE may be configured both with intra-CU and inter-CU LTM candidate cells.

To support inter-CU LTM over the Xn interface, 3GPP decided to reuse the existing inter-node messages defined in [12]. The Xn signaling is enhanced by introducing support for inter-CU candidate configuration in *HANDOVER REQUEST* message and (early) data forwarding *EARLY SN STATUS TRANSFER* message. Also, new Xn interface messages, which are counterparts of Rel. 18 F1 interface messages, are introduced to support enhanced inter-CU LTM features. More specifically, the *TA INFORMATION TRANSFER* message is used to transfer the estimated TA from the target CU to the source CU. To facilitate the separation of mobility preparation and potentially RACH-less execution, a *CELL SWITCH NOTIFICATION* message was introduced to inform the target CU about the upcoming UE access to the target cell. Finally, subsequent mobility is supported using the *LTM CONFIGURATION UPDATE* message, which provides LTM configurations to be used by candidate network nodes (gNBs) when they start serving the UE.

2) L2 RESET, SECURITY KEY UPDATE AND RECOVERY

In the baseline handover, the determination of L2 reset behavior at the UE, more specifically, for the Radio Link

Control (RLC) and Packet Data Convergence Protocol (PDCP) layers, is based on flags included in the radio bearers configuration in the RRC reconfiguration message triggering the handover (*RRCReconfiguration* message), namely *reestablishRLC*, *reestablishPDCP*, which is used for data radio bearers (DRBs), and *discardOnPDCP*, which is used for signalling radio bearers (SRBs). Avoiding re-establishment theoretically allows the UE to maintain high throughput during mobility. For DRBs, in Rel. 18 intra-CU LTM, where PDCP reestablishment is not needed, an ID (*ltm-NoResetID*) was introduced to allow the UE to determine whether RLC reestablishment is needed, not only upon the initial, but also upon subsequent cell switches. If the value of *ltm-NoResetID* parameter for both the source and target cell is the same (which effectively indicates that the two cells are provided by the same distributed unit (DU), and, consequently, the same RLC entity on the network side), the UE does not reestablish the RLC entity. If the IDs are different, RLC reestablishment and PDCP recovery, i.e., retransmission of PDCP packets previously submitted to RLC layer but pending successful delivery confirmation, are performed for RLC bearers configured with acknowledged mode (AM). For SRBs, L2 reset behavior is determined by the *reestablishRLC* and *discardOnPDCP* flags, as in the baseline handover.

A similar mechanism was introduced in Rel. 19 to distinguish inter-CU cell switch, where a security key update is needed, in contrast to the intra-CU cell switch, where a security key update is not necessary. If the value of the *noSecurityChangeID* field for the source and the target cell is different, it indicates an inter-CU cell switch. In this case, the UE derives a new security key using the 3-bit next-hop chaining count (NCC) value provided in the Rel. 19 enhanced cell switch command (CSC), reestablishes the PDCP and RLC layers for all radio bearers, and uses the derived key for PDCP packets (de)ciphering. If the *noSecurityChangeIDs* are the same, indicating an intra-CU cell switch, the UE follows the Rel. 18 behavior for DRBs, while for SRBs the UE discards the stored PDCP data units and reestablishes RLC. Further details on security-related aspects are provided in Section II-B.

It should be noted that the presence of the NCC value in the CSC may serve as an implicit indication of whether security key update and PDCP reestablishment is needed, which at first glance seems to make the introduction of *noSecurityChangeIDs* unnecessary. Nevertheless, *noSecurityChangeIDs* are still needed to determine the execution of LTM recovery after a potential failure event, namely radio link failure (RLF) or handover failure (HOF). More specifically, in case an RLF occurs while the UE is configured with LTM, the UE performs cell selection and if the selected cell is configured as an LTM candidate and its *noSecurityChangeID* is the same as *ltm-ServingCellNoSecurityChangeID* (The information element (IE) stores the *NoSecurityChangeID* of the source cell.) the UE avoids the conventional RRC reestablishment procedure and performs LTM recovery

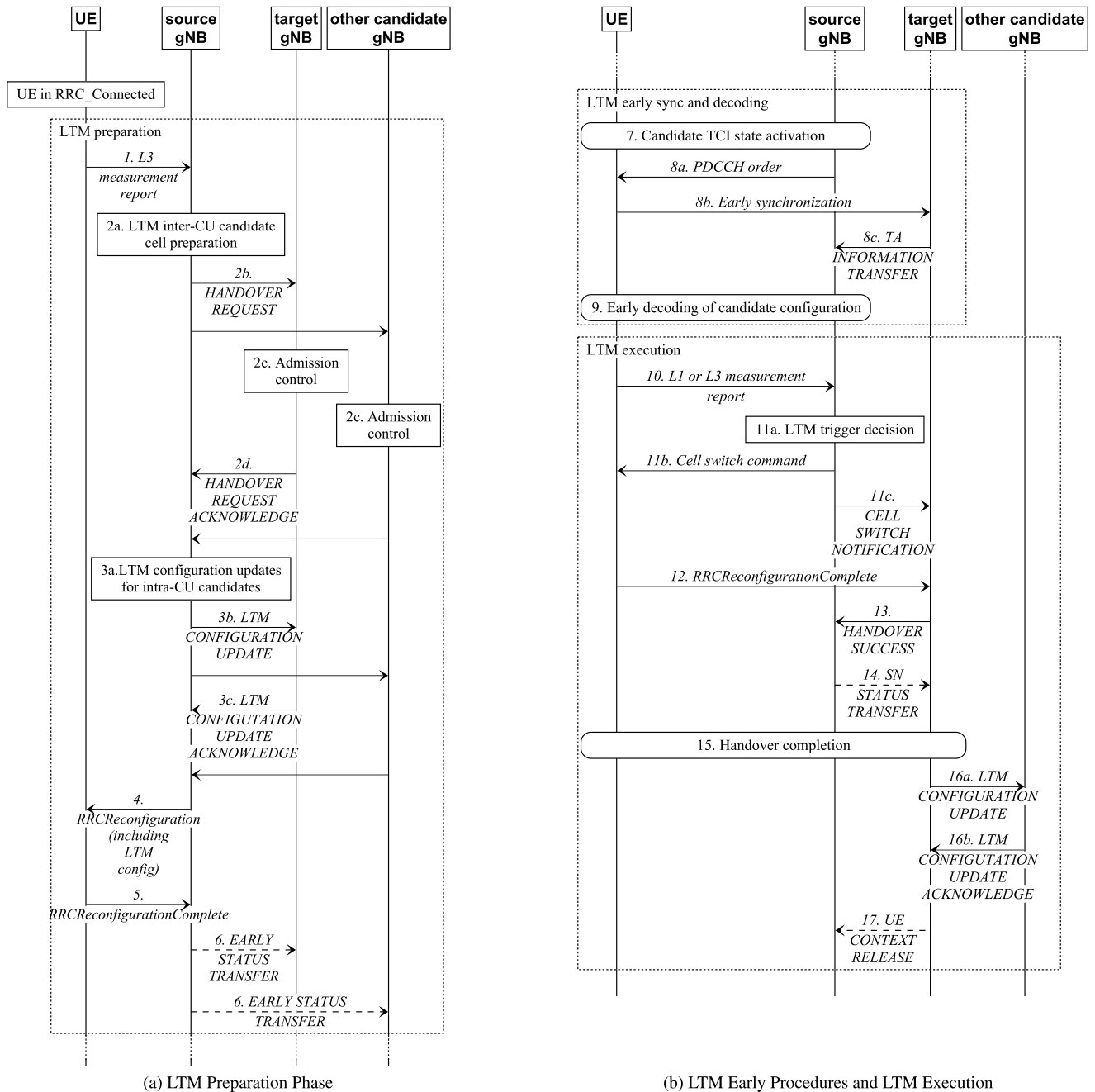


FIGURE 1. Inter-CU LTM procedure taken from [11].

by directly applying the candidate configuration of the selected cell. In case of HOF, the procedure is similar, with the difference being that, to perform LTM recovery, the *noSecurityChangeID* of the selected cell must be the same as that of the failed handover target.

3) CSI RESOURCE CONFIGURATION

L1 measurement and reporting is based on the configuration of CSI resource sets, each of which indicates a set of reference signals (RSs), potentially from multiple cells,

to be measured by the UE. The resource sets are linked to report configurations, which are serving cell-specific, i.e., associated with a specific cell and used when this cell is serving, and determine how and when the UE reports measurements in the associated resource sets. CSI resource configuration in Rel. 19 LTM is more intricate compared to Rel. 18 due to:

- the need for coordination of CSI resources among multiple CUs

- the support of CSI-RS based L1-RSRP measurement and reporting. This is on top of Rel. 18 synchronization signal blocks (SSBs) reporting.
- CSI-RS-based early CSI acquisition. Obtaining finer information of SSBs candidate cell for appropriate link adaptation in terms of channel quality indicator (CQI), rank indicator (RI), and precoding channel matrix (PMI).

The first of these challenges raises the question of which entity should be the one configuring the resource sets. The proper definition of the resource sets is not a straightforward task, as site-specific knowledge may be used for optimization of their content, with the aim of minimizing the measurement and reporting overhead of the UE, while ensuring that mobility-relevant measurements are not omitted. The Rel. 19's decision to make source CU responsible for constructing the CSI resource sets and providing them to the candidate CUs was to avoid the above complications and to avoid excessive specification effort.

Regarding the second challenge, i.e., the support of CSI-RS based measurements, the issue is that, unlike SSB configurations of candidate CUs, which are known to the source CU upon the setup of the Xn link between them, the CSI-RSs configurations, which can be UE-specific, are not known to the source CU. Therefore, for the configuration of the CSI resource sets based on CSI-RSs, additional signaling within the UE context setup is required for the source CU to collect the CSI-RS resource configurations from the candidate CUs, both for L1-RSRP measurements and early CSI acquisition. More specifically, a request for providing CSI-RS resource configuration, indicating also the purpose, i.e., L1-RSRP measurement or CSI acquisition, is included in the *HANDOVER REQUEST* (step 2b in Fig. 1) and/or *UE CONTEXT SETUP REQUEST* ((candidate) CU to candidate DU – intra-CU signaling not shown in Fig. 1) and the candidate CU and/or the candidate DU responds by providing the requested configuration. For early CSI acquisition, the candidate DU may also provide the configuration of channel state information interference management (CSI-IM) resources, in addition to CSI-RS resources, for interference measurements. After collecting the CSI-RS (and optionally CSI-IM) configurations, the source CU can prepare the common CSI resource sets, which may include resource sets based on SSBs, CSI-RS resources for L1-RSRP measurement, and CSI-RSs (and CSI-IM) resources for CSI acquisition. The resulting common resource sets are then used for CSI reporting configurations for both L1-RSRP measurements and CSI acquisition. The source CU shares the common CSI resource configuration with the source DU to enable collection of L1-RSRP measurement reports. It also shares the common CSI resource sets with the candidate CU and/or candidate DU through an *LTM CONFIGURATION UPDATE* and/or *UE CONTEXT MODIFICATION REQUEST*. The candidate DU can subsequently provide reporting configurations for

its candidate cells for RSRP measurements (to be used in a potential subsequent cell switch when the UE moves to one of those candidate cells), and/or reporting configuration, maximum up to one reporting configuration for each of its candidate cells, for early CSI acquisition.

4) REFERENCE CONFIGURATION

Another novelty of Rel. 18 LTM was the use of a reference configuration, which allows the candidate cell configurations to be defined as a delta over the common reference, enabling signaling overhead reduction and efficient target cell configuration application. The reference configuration leverages the anticipated similarities of the configurations of candidates under the same CU.

The extension to an inter-CU setup presented two design choices, namely to either follow Rel. 18 in defining a single reference configuration for all CUs or to define a per-CU reference configuration. The former choice, which was the one selected, results in lower specification effort and lower signaling overhead for reference configurations but can lead to potentially higher signaling impact for the individual candidate configurations in terms of their size, as there is less room for similarity exploitation among cells belonging to different CUs.

B. SECURITY ISSUES

This section introduces the security background and the security issue that is novel to Rel. 19 as Rel. 19 work introduces inter-CU LTM. In Rel. 18 assumption has been that the LTM procedure runs as intra-gNB without any possibility of having a security update during the LTM cell switch procedure. To explain the necessary security know-how, we introduce the background of key change in baseline handover of 5G.

1) SECURITY FUNDAMENTALS BACKGROUND

Key generation for UE has its own complexities. We will try to explain it briefly as introduced in Fig. 2 for the enhancements of LTM procedure. For each connected UE, each RAN node will generate and share its own key with the UE. The key from RAN node is built in coordination with AMF. This is best described in Fig. 2, which is from [13]. Each key for gNB (KgNB) represents a new RAN node key. The initial KgNB, marked as initial in the figure, is generated from the AMF key access management function (KAMF) using *UL NAS COUNT*. After the initial KgNB is created, the next key can be generated horizontally or vertically, as shown with vertical and horizontal lines in the figure. In case horizontal key generation is used, only the values of physical cell ID (PCI) and DL frequency are used as input along with the last KgNB to create the new key. However, if the vertical key generation is used, a new parameter next hop (NH) is added to the key generation, which is a parameter calculated by AMF and indicated to the RAN node to generate the next key. The NH ensures that the security domain is separated

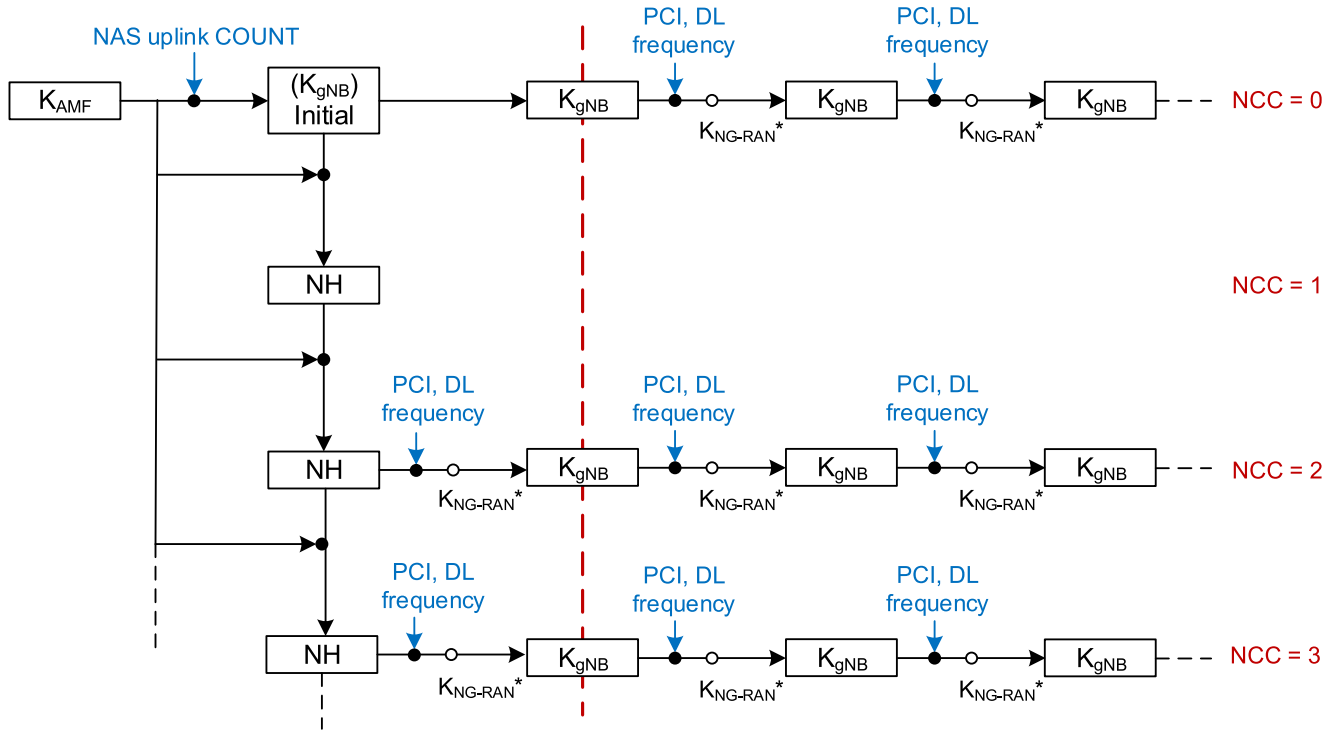


FIGURE 2. Illustration of handover KgNB key chaining from [13].

between gNBs, i.e., the possession of the latest key is not sufficient to generate the next key for the UE.

2) SECURITY CONFIGURATION

In the previous section, we explained the horizontal and vertical key generation. In this section, we explain the configuration aspects of this. The key generation is done independently at the UE side and network side to avoid the transmission of the key over the radio. Thus, only the necessary parameters, such as the NCC value, are indicated to the UE such that the UE will generate the key as the network expects. On the right-hand side of the Fig. 2, there are NCC values listed for each row. Each row represents a vertically generated key chain and the isolation of the chain. The NCC parameter indicates to UE whether it shall do a horizontal or vertical key derivation. The IE *MasterKeyUpdate* is used to indicate the key change parameters to the UE. In case the NCC value stays the same as the last key, UE knows it will do horizontal key generation. In case the NCC value is increased, the UE generates the NH value by itself through the security algorithm configured by AMF at initial security setup. UE then uses this value to generate the KgNB. Similarly, in the security configuration, security algorithms can be updated or KAMF can be changed, but as this was not in the scope of Rel. 19 LTM work.

3) SECURITY DURING HANDOVER

Technical specification for security architecture and procedures for 5G system [13] discusses three types of handovers for key handling during handover.

- Intra-gNB handover
- Xn handover
- N2-handover

This paper will not discuss N2-handover as NG-based LTM is not part of the Rel. 19 scope. During handover, the *PATH SWITCH REQUEST* (contacting AMF for the first time) sent after handover completion dictates which of the vertical or horizontal keying is used. In both intra-gNB and Xn handover, the source gNB is responsible for generating the target key and sharing the key with the target gNB. To generate the target key, source gNB uses the stored parameters indicated by the AMF in the latest path switch or initial UE message. The *PATH SWITCH REQUEST ACK* from AMF, following a previous handover, indicates NCC value and NH value to the gNB for it to store and use in the next key generation. If the source gNB has a fresh NH value, then it will generate a vertical key. Otherwise, it will generate a horizontal key.

One example is illustrated in Fig. 3 where source gNB has a fresh NH value stored from the previous path switch procedure. After the measurement report source determines to handover the UE. It generates the KgNB* for the target gNB and does this with vertical key generation. Then the source sends the key and NCC value in the *HANDOVER REQUEST* to the target node. Target node sees that it is a vertical key generation and indicates the incremented NCC value to the UE in the handover command (the *RRCReconfiguration* message). The UE receives the handover command from the source gNB and decodes the

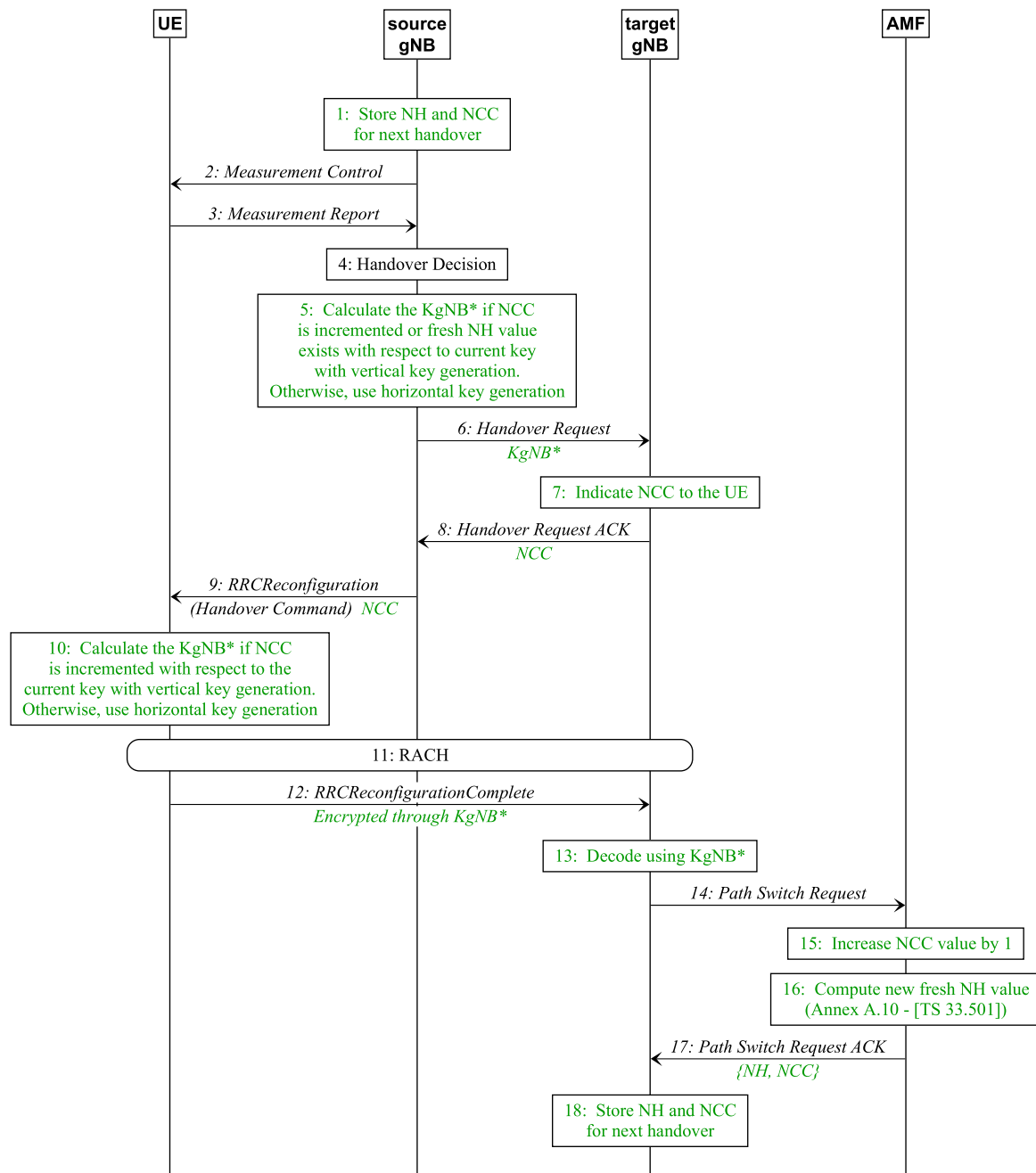


FIGURE 3. Example message sequence chart for key chaining during an inter-gNB (inter-CU) HO. Green font indicates information related to key generation.

target configuration and observes that the NCC value in *MasterKeyUpdate* IE is incremented, which indicates vertical key generation. Then, UE generates the new NH value and uses PCI and DL frequency to generate KgNB*. UE sends the *RRCReconfigurationComplete* encrypted with KgNB* to target gNB. The target gNB can decode the message as it has received the key from the source gNB. Then the target node sends *PATH SWITCH REQUEST* to the AMF. AMF increments the NCC value and computes the fresh

NH value and indicates the pair to the target gNB in *PATH SWITCH REQUEST ACK*. Target gNB stores the NH and NCC value. At this point, the next steps are left to RAN node implementation. The source gNB knows the security key of the target gNB for the UE. Thus, the target gNB can send another *RRCReconfiguration* and update the security key of the UE using another vertical key generation. This would separate the security domains of source and target gNB. However, this would require an intra-cell handover,

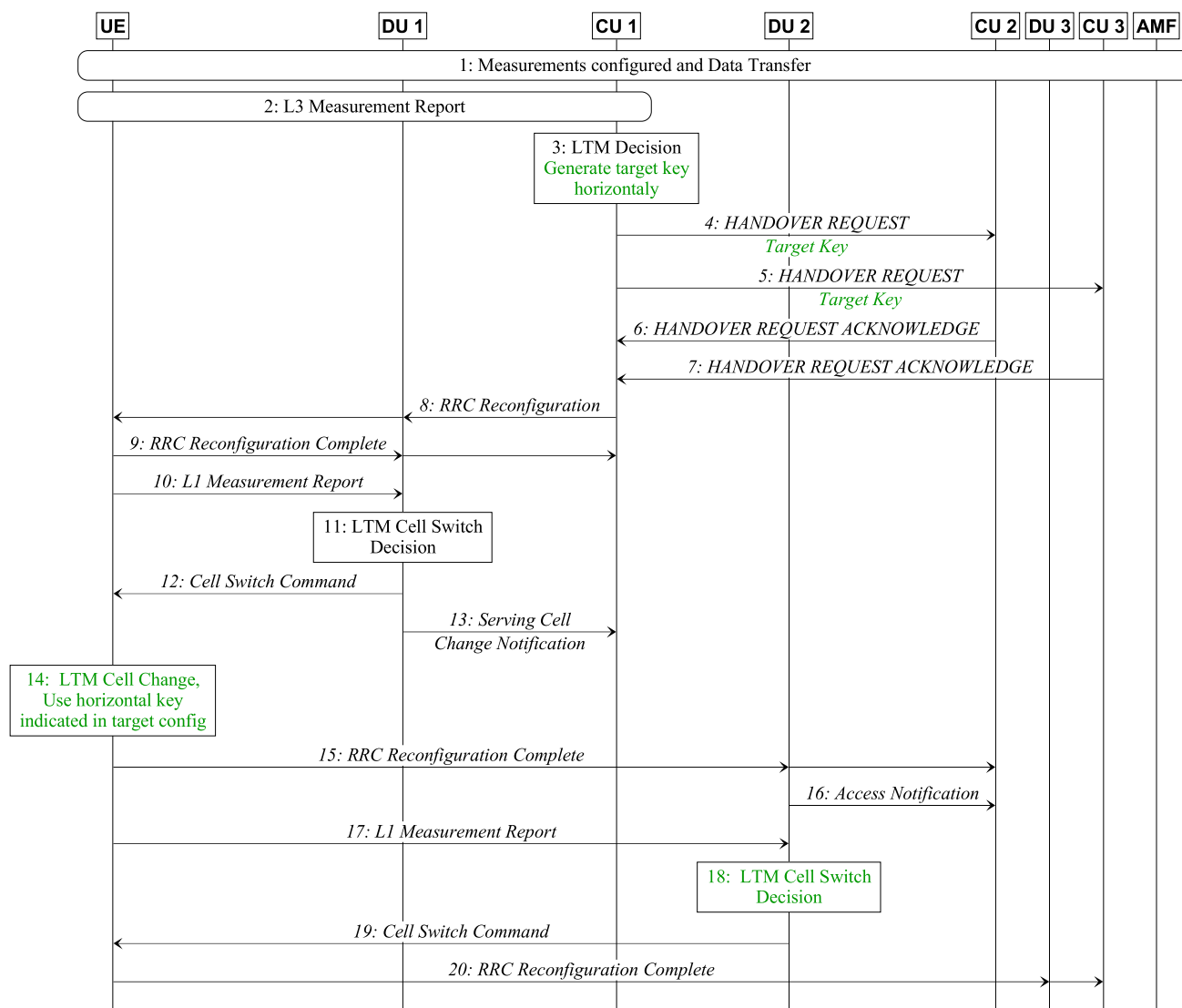


FIGURE 4. Example of Inter-CU LTM procedure (without Rel. 19 enhancements on security) to indicate how would the key generation work at the network and the UE side for the first cell change. Green font indicates information related to key generation.

which would cause interruptions. It is also possible to wait until next handover and keep using the KgNB* generated by the source.

4) SECURITY FOR SUBSEQUENT LTM

In order to demonstrate the security problem that happens for LTM, we start with the assumption that the *MasterKeyUpdate* IE can go in candidate configuration. During initial preparation, the source gNB indicates the key to the target gNBs, and the candidate configurations are built, and the *MasterKeyUpdate* value is fixed in the candidate configuration, irrespective of whether the NCC value is kept the same or incremented. UE receives all the candidate configurations in steps 8-9 of Fig. 4. The source DU may trigger a cell switch towards one of the candidate cells, and UE will decode the candidate configuration and can send the

RRCReconfigurationComplete to the target cell. UE uses the *MasterKeyUpdate* IE to generate the new key. The first cell switch works without a problem. However, for subsequent cell switches, there are three issues:

- 1) The path switch procedure from the target node may not be completed. This creates a race condition between the security context at AMF and the UE.
- 2) The UE is indicated with the candidate configuration with a fixed NCC value indicated in the *MasterKeyUpdate* IE, which may not be accurate depending on the previous handover. UE will keep generating horizontal keys in subsequent handovers as it re-uses the configuration, including the same NCC value.
- 3) The target gNB is indicated with a key at the preparation, which is not valid anymore as it was built

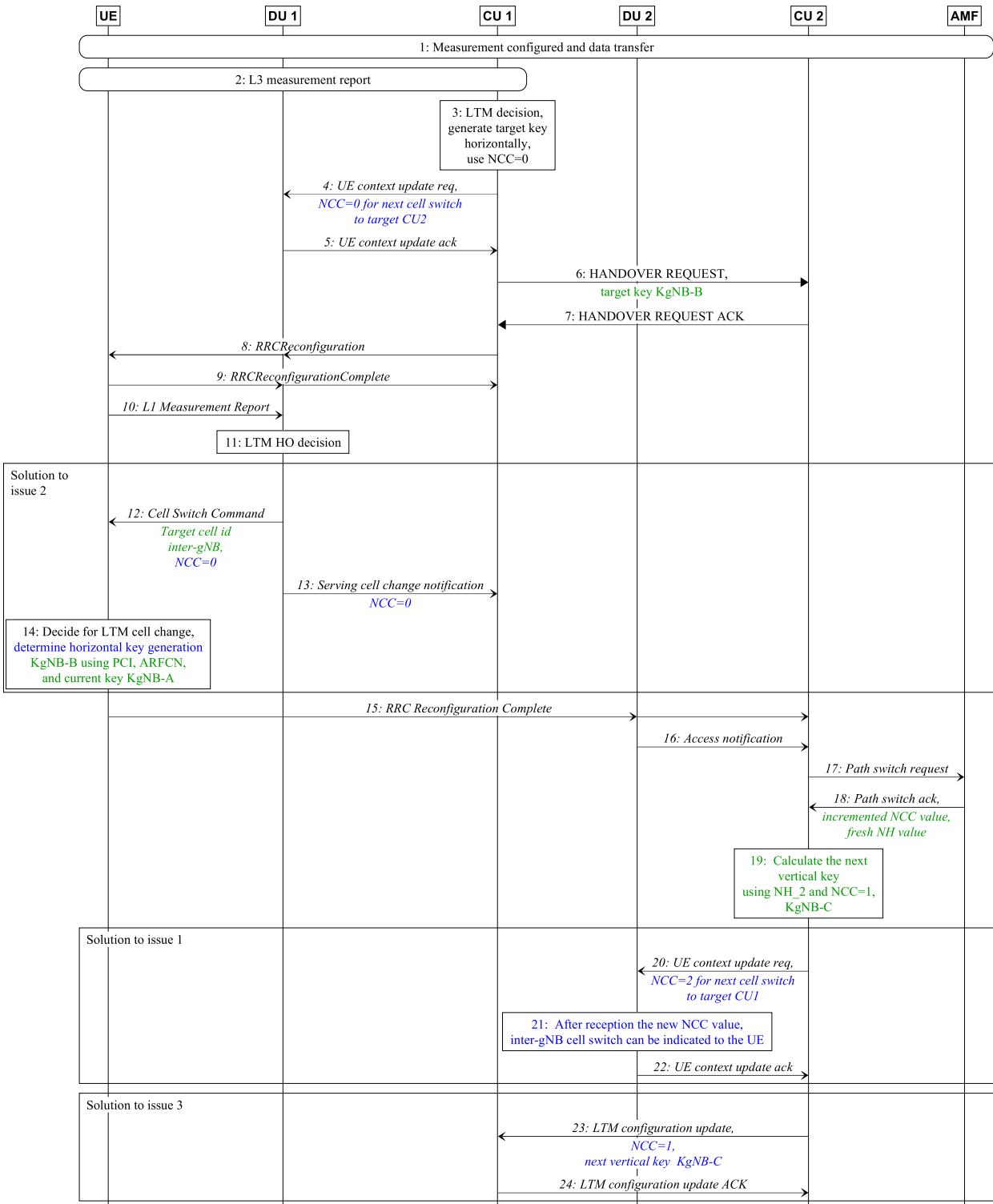


FIGURE 5. Inter-CU LTM cell switch procedure as an example involving a horizontal key update. Boxes captures the changes introduced to support subsequent mobility that enables a key update without RRC signalling. Green font indicates information related to key generation. Blue font indicates novel behavior specified in Rel. 19.

based on the source gNB key. But the next key shall be based on the current source, which may be different than the initial source gNB.

In order to solve these three issues, 3GPP has introduced the following modifications to the security procedure in Rel. 19 LTM.

- 1) The source DU waits for receiving the NCC value from source CU before indicating CSC to the UE to avoid the race condition between AMF and the DU. (Steps 21-23 in Fig. 5.)
- 2) The medium access control control element (MAC CE) CSC is enhanced to indicate the NCC value dynamically to the UE, and the MasterKeyUpdate IE shall not be configured in candidate configurations. (Steps 12-14 in Fig. 5.)
- 3) A new Xn message is introduced to update the keys for candidate cells after each handover completion. (Steps 23-24 in Fig. 5.)

For a short analysis of vulnerabilities related to this procedure we would like to underline that the MAC CE is not an integrity protected or encrypted message. From this regard an attacker can send a MAC CE cell switch message including a malicious NCC value to lead UE to handover failure. The use of malicious NCC value will result in UE generating a key that may not be the same as the key expected by the network, as such the *RRCReconfigurationComplete* may not be decoded by the network. However, this is a robustness issue but not security issue as the attacker cannot know the NCC of the UE as the NCC value is used to generate the next key depending on the current key. As long as current key is not compromised this shall not be a security issue.

C. DC ASPECTS

When the UE is in DC, it is served by an MCG, which includes the primary cell (PCell), and an SCG, which includes the primary secondary cell (PSCell). Already in Rel. 18, the UE can be configured with intra-CU MCG LTM and/or intra-CU SCG LTM independently, without the cell switch in one cell group affecting the configurations in the other. Nevertheless, inter-CU cell switches in either cell group may impact the configuration of the other. In this section, we have a closer look at the enhancements introduced in Rel. 19 to support inter-CU LTM cell switches when the UE is in DC. LTM with inter-CU candidates cannot be configured simultaneously in both cell groups; in other words, when inter-CU LTM is configured in one cell group, LTM may be simultaneously configured in the other cell group, but including only intra-CU candidates.

1) INTER-CU MCG LTM

The procedure for inter-CU MCG LTM when UE is in DC is very similar to that described in Section II-A, with some additional considerations for the impact on the SCG. More specifically, the candidate configurations may include updates of the SCG configuration, including PSCell change, release of the SCG, or even addition of SCG.

As discussed in Section II-A, an inter-CU PCell change results in a security key update. Furthermore, an update of the security key of the SCG is required, as the secondary node (SN) key is derived using the security key of the master node (MN) and a SN-specific counter value. When

applying the target configuration, the UE releases the existing SCG configuration and applies the new SCG configuration, performing also (DL and UL) synchronization with the SN. We note that the early sync procedures are not applicable for synchronization with the SN in inter-CU MCG LTM. Since security keys are updated in both cell groups when the *noSecurityChangeIDs* of the source and the target configuration are different, the PDCP and RLC layers of all data radio bearers in both cell groups must be reestablished.

2) INTER-CU SCG LTM

The procedure for inter-CU SCG LTM has a few distinctive characteristics compared to inter-CU MCG LTM. The underlying network topology is not a full mesh, but a star, with the MN having an Xn link with the source and each of the candidate SNs. The source SN initiates the procedure by sending an *SN CHANGE REQUEST* to the MN, indicating a list of candidate PSCells, potentially accompanied by a set of measurements for candidate cells, and including CSI resource request and early synchronization configurations, which are forwarded to the respective SNs.

The MN requests and receives the CSI resource configurations and early synchronization configurations from the candidate SNs for the accepted candidate PSCells, which it forwards to the source SN, along with the SCG part of the candidates' RRC configurations. The SN replies with the common CSI resource configuration and its updated SCG configuration. With an SN modification request exchange, the MN forwards the CSI resource configuration and the early synchronization configurations of candidate cells to each SN and receives updated SCG configurations from them.

Alongside the early synchronization configurations, the resource configuration is provided by each candidate SN to form the SCG configuration. The actual use of the resources provided is only after successful completion of the inter-SN LTM procedure. This is aligned between the MN and SN through receiving the *RRCReconfigurationComplete* message by MN and the same message being forwarded to SN by MN.

The MN has all the information required for generating the LTM configuration. Each candidate RRC configuration must also contain an MCG part, which is provided by the MN. A reference configuration can be used, with the SCG part of the reference configuration being provided by the source SN or any of the candidate SNs upon request, and the MCG part, if any, provided by the MN. The MN ensures that the candidate RRC configurations are complete configurations, potentially resulting from combining the reference configuration with a delta configuration provided by the SNs.

Although it is evident that the MN plays a central role in the procedure, it is the source SN that not only triggers the preparation, but also the early sync procedures and the cell switch. Upon reception of the CSC, the UE applies the target configuration and sends an *MN RRCReconfigurationComplete* message, which includes an

SN RRCReconfigurationComplete message to the MN that is forwarded by the MN to the SN.

The L2 reset and security key update procedure is very similar to that specified in Rel. 18 for the subsequent conditional PSCell addition or change (S-CPAC). A *noSecurityChangeID* is assigned to each candidate configuration, and a corresponding sk-counter list is provided to the UE, which, as discussed above, is used together with the master key to derive the secondary key. We note that a list per *noSecurityChangeID*, i.e., per SN, is provided to the UE to ensure that every time the UE returns to an SN, e.g., after an SN change, it derives a different security key by using the first unused sk-counter value from the list corresponding to that SN.

The L2 reset behavior is largely determined by comparing the source and target *noSecurityChangeID*, but in a DC scenario, there can be some bearer configurations that require more detailed treatment. To this end, it is useful to introduce the notion of the termination point of a radio bearer, which can be either the MN or the SN, and it indicates where the PDCP entity of the radio bearer lies. For example, an MN-terminated SCG DRB may be configured, whose RLC and lower layers make use of SCG resources, but the PDCP layer is at the MN. Consequently, an SN security key update does not imply PDCP reestablishment for such a bearer. For SN-terminated DRBs or DRBs that changes termination point (convert from MN to SN terminated or vice-versa), a security key update is required, and the PDCP and RLC entities are reestablished. For SRB3, PDCP reestablishment is also determined based on the L2 reset behavior is largely determined by comparing the source and target *noSecurityChangeID*, while for SRB1 and SRB2, the L2 reset behavior is determined by flags in the target configuration, as discussed in Section II-A2.

D. MEASUREMENT ENHANCEMENTS

The measurement framework in Rel. 19 LTM has been significantly enhanced to address efficiency and accuracy concerns identified in earlier releases. These enhancements focus on enabling narrow beam based cell switching through CSI-RS-based measurements and reducing unnecessary measurement reporting overhead through event-based reporting mechanisms. The improvements also include better integration with existing RRC measurement procedures and enhanced support for various deployment scenarios. These measurement enhancements are crucial for ensuring that LTM can operate efficiently in diverse network conditions while maintaining the required mobility performance.

1) CSI-RS BASED MEASUREMENTS

In Rel. 18, only SSB-based L1-RSRP measurements for candidate cells were supported. These measurements were primarily used to trigger the early synchronization procedure and initiate cell switching, using a SSB based target beam, at the source node. Additionally, based on these measurements, Rel. 18 also supported tracking reference

signal (TRS) based beam selection to enable fine time and frequency synchronization with the target cell [1].

In Rel. 19, the framework was extended to support CSI-RS resource-based L1-RSRP measurements. This enhancement enables the selection of narrower CSI-RS resource-based beams in the spatial domain, compared to SSB-based beams, from the target cell for cell switching. As part of this extension, Rel. 19 allows a candidate TCI state to be associated with a CSI-RS resource for spatial domain (i.e., quasi-co-location Type-D), and a TRS resource for fine time and frequency synchronization (i.e., quasi-co-location Type-A). This capability is mainly beneficial for FR2 deployments, where high propagation loss motivates the use of narrower beams to improve coverage.

To enable CSI-RS resource-based L1-RSRP measurements, the UE can be configured with network-triggered reporting (e.g., periodic, semi-persistent, or aperiodic) and/or event-triggered reporting, associated with a set of CSI-RS resources from candidate cells.¹ The reporting configurations are provided under the current serving cell, similar to Rel. 18 reporting configurations for SSB-based measurements. As with SSBs, a measurement set associated with a report may contain CSI-RS resources from one or more candidate cells, provided under one of the LTM CSI resource configurations, which are configured commonly across all candidate cells. The configuration of each CSI-RS resource for a candidate cell is provided in the LTM configuration of the respective candidate. This information is placed outside the candidate cell's RRC Reconfiguration, allowing the UE to access it without decoding the candidate cell's RRC configuration.

Both periodic and semi-persistent (SP) CSI-RS resources are supported for network-triggered reporting. A report can be associated with either a set of periodic CSI-RS resources or a set of SP CSI-RS resources from one or more candidate cells. An explicit indication is provided for each measurement set to identify the resource type (periodic or SP) of the CSI-RS resources in that set. The configuration parameters, such as time and frequency allocation, associated with a CSI-RS resource follow the framework that has been used since new radio (NR) Rel. 15.

Periodic CSI-RS resources are assumed to be active after the RRC configuration containing the LTM configuration is provided to the UE. SP CSI-RS resources are selected dynamically by the source node, for example, based on other measurements such as SSB-based L1 measurements or periodic CSI-RS resource-based measurement results. To activate the selected SP CSI-RS resources from a candidate cell, the source cell first sends an activation request, i.e., *CSI-RS COORDINATION REQUEST*, to the candidate cell via their respective DU and CU coordination signaling, including the information of the SP CSI-RS resources to be

¹Note that in this subsection, the focus is on network-triggered reporting, while a detailed discussion of event-triggered reporting is provided in Section II-D2.

activated [12], [14]. Upon receiving confirmation from the candidate cell, i.e., *CSI-RS COORDINATION RESPONSE*, the source cell sends a MAC CE, i.e., SP CSI-RS/CSI-IM Resource Set Activation/Deactivation for Candidate Cell MAC CE [15], to the UE to indicate activation of the SP CSI-RS resources, after which the UE may begin measurements. The same procedure is used to deactivate a previously activated SP CSI-RS resource. To obtain measurements from activated SP CSI-RS resources, the source node can either trigger an SP report via MAC CE or downlink control information (DCI), or trigger an aperiodic report via DCI, in which the activated SP CSI-RS resources are configured as the measurement RSs for the SP and aperiodic CSI-RS reporting configuration [16].

For CSI-RS resource-based measurement reports, the UE reports a configured number of CSI-RS resource indicators and their corresponding RSRP values. The report format for a network-triggered reporting is identical to the Rel. 18 network-triggered SSB-based measurement report, except that a CSI-RS resource indicator is used in place of an SSB resource indicator [17]. As with SSB-based measurements, for network-triggered reporting, the UE can be explicitly configured to report measurements for the current serving cell by enabling *SpCellInclusion* IE. This is particularly relevant when the serving cell is also configured as a candidate cell and its CSI-RS resources are included in the measurement set. To determine the CSI-RS resources in the measurement set corresponding to the current serving cell, the UE uses the PCI of the associated candidate cell and the frequency information of the SSB quasi co-located with the CSI-RS resources (from SSB-Frequency), i.e., CSI-RS resources in the measurement set are considered to correspond to the current serving cell if both the PCI and the center frequency of the cell-defining SSB match those of the current serving cell [16].

From the UE requirements point of view, only CSI-RS resource-based intra-frequency L1-RSRP measurements are supported in Rel. 19 [18]. A measurement is defined as a CSI-RS resource-based intra-frequency measurement provided that at least 48 physical resource blocks (PRBs) of the CSI-RS resource of LTM candidate cell(s) configured for L1 measurement is included within the current active DL band-width part (BWP) of the UE, and if the sub-carrier spacing (SCS) and cyclic prefix (CP) type of the CSI-RS resource are aligned with those of the DL BWP. Otherwise, a measurement is CSI-RS resource-based inter-frequency measurement.

To determine the receive beam for a CSI-RS resource, two options are considered. One option is to use the receive beam determined for the SSB, which is quasi co-located with the CSI-RS resource. Another option is to allow the UE to perform beam sweeping to determine the receive beam corresponding to the CSI-RS resource, which may require additional delay, and this has been reflected in the CSI-RS resource-based L1 measurement delay requirements [18].

During RAN1 Working Group discussions, the potential support of L1-signal-to-noise-plus-interference ratio (SINR) measurements using CSI-RS resources was also considered. Many operators expressed interest in L1-SINR as a useful quantity for assessing channel interference conditions and identifying overloaded frequencies, enabling the network to hand over UEs to more lightly loaded parts of the spectrum. However, concerns were raised regarding the intermittent nature of interference and the lack of agreed-upon measurement filtering. As a result, no consensus was reached to support L1-SINR in Rel. 19. Nevertheless, this topic may re-emerge for consideration in future releases.

2) EVENT TRIGGERED REPORTING

According to Rel. 18, measurement reporting for LTM can be configured as network-triggered periodic, SP, or aperiodic reporting. Among these, periodic reporting is the one that does not require any additional triggering from the network. However, in such a type of reporting, irrespective of the signal's quality for the serving or neighbor cell, the measurement reports are always transmitted from the UE to the gNB. Such an approach might possibly result in resource and signaling inefficiency as the gNB is notified by the UE even if the quality of the radio link is still acceptable. Furthermore, it may also lead to the lack of on-time measurement reporting, i.e., when the radio link between the UE and the gNB becomes weak, the UE waits for the next reporting period to deliver the instantaneous measurement results instead of doing it immediately.

Due to the above-mentioned reasons, one of the objectives of 3GPP Rel. 19 WI on enhancing LTM [10] was to enable event-triggered L1 measurement reporting. Another motivation was to align L1 measurement reporting for LTM with L3 measurement reporting for the legacy handover procedure, where both periodic and event-triggered reporting is supported. Namely, the L1 equivalents of the L3 measurement events A1 – A6 [19] were specified with the intention to trigger the measurement reporting once an individual beam's RSRP becomes higher or lower than the threshold or when it becomes offset better than the beam used for comparison, e.g., serving beam. The full list of L1 measurement reporting events is as follows [15]:

- Event LTM2: Beam of serving cell becomes worse than the threshold
- Event LTM3: Beam of candidate cell becomes offset better than the beam of the serving cell
- Event LTM4: Beam of candidate cell becomes better than absolute threshold
- Event LTM5: Beam of serving cell becomes worse than threshold1 and Beam of candidate cell becomes better than threshold2

For an event-triggered reporting, i.e., an LTM CSI report, configured with report type set to *eventTriggered*, candidate beam measurements are performed using the RSs, either

SSBs or CSI-RS resources, configured in the LTM CSI resource configuration associated with the LTM CSI report. The serving beam measurement is performed using the RS configured in the indicated TCI state, or an RS that is quasi-co-located with the RS configured in the indicated TCI state in the serving cell [16], with the same RS type (SSB or CSI-RS resource) as that associated with the candidate beam.

The introduced LTM events provide a complete set of tools to monitor either when the serving cell's beam becomes weak (LTM2), the candidate cell becomes sufficiently strong (LTM4), or to compare directly the serving's beam with the candidate's beam (LTM3). It is obviously possible to configure the UE with multiple such events aimed at measurements for different cells, beams, or other purposes, such as LTM cell switch or early synchronization towards one of the configured candidate LTM cells. The entering conditions for each of these events need to be fulfilled for the duration of time-to-trigger (TTT) – a configurable period controlling the stability of the measurement result used for deciding whether the measurement report (MR) shall be sent by the UE. The UE can also be instructed to transmit a MR when the leaving condition for any of the events has been met (i.e., report on leave). When the MR is triggered, the UE is supposed to fill the MAC CE with measurement results, according to the format defined in [15]. MR MAC CE comprises reference signal resource index (RSRI) that represents both SSBs and CSI-RS resources, for each reported beam and a corresponding RSRP value. In addition, each reported beam has its type. This field is used to indicate if the beam is reported as it has met the entry or leaving criteria for the specific LTM measurement event, or has met the event previously and is being re-reported, or did not meet any event. The last octet of the MR MAC CE is used for conveying the RSRP value of the serving beam. [15] also defines a truncated format of the MR MAC CE, which shall be used by the UE when the available UL grant is not sufficient to accommodate the full MR. UE has to request another UL grant in cases where the truncated MR cannot fit to the available UL grant.

Additionally, similar to L3 event-triggered reporting, periodic reporting can also be optionally configured for an event-triggered L1 measurement report. In this case, after the first event-triggered report, the UE is configured to send reports periodically with a configured interval and for a configured number of times. This mechanism can provide the network with additional details on the variation in the quality of the triggered beams.

3) L3 MEASUREMENT BASED LTM

In the later phase of Rel. 18 discussions on LTM, it was agreed that the LTM procedure may also be triggered based on L3 measurements reported by the UE. Specifically for frequency range 1 (FR1), it was decided that the LTM procedure can be executed without relying on L1 intra- or inter-frequency measurements. This may occur when the UE does not indicate support for L1 measurement and reporting for candidate

TABLE 1. Mobility trigger information for CU-DU mobility initiation request.

Information Element	Description
Triggering Indication	- Indicates the type(s) of procedure to be triggered: -- Early UL synchronization -- Early DL synchronization -- Cell switch
Cell Switch Information (Included when cell switch is indicated as the Triggering Indication)	Indicates identity of the target cell selected by the CU and associated beam information, such as the SSB index corresponding to the TCI state(s) to be indicated in the cell switch command.
Early UL Synch Information (Included when Early UL synchronization is indicated as the Triggering Indication)	- Includes: -- A list of candidate cells -- For each candidate cell, a list of associated beams (i.e., SSB indices) corresponding to SSBs to be indicated in the PDCCH orders for early UL synchronization.
Early DL Synch Information (Included when DL synchronization is indicated as the Triggering Indication)	- Includes: -- A list of candidate cells -- For each candidate cell, a list of associated beams (i.e., SSB indices) corresponding to TCI states to be indicated in the early TCI state activation commands for early DL synchronization.

cells, and/or when the network chooses not to configure L1 measurements. In such cases, L3 measurements reported by the UE serve as a fallback mechanism. This implies that the source cell must rely solely on L3 measurements to make decisions regarding early DL synchronization, early UL synchronization, or cell switching. This may be achieved by configuring the UE to report beam-level information within the L3 measurement reports. This design choice to allow L3-based LTM for FR1 was finalized in the RAN4 working group during the later stages of Rel. 18. As a result, relevant discussions in other working groups, particularly RAN3, which is responsible for inter-node signaling over interfaces such as F1 and Xn, did not occur at the time of our earlier work [1]. Since this work aims to provide a comprehensive overview of the enhancements made to the LTM framework after our previous work, we now

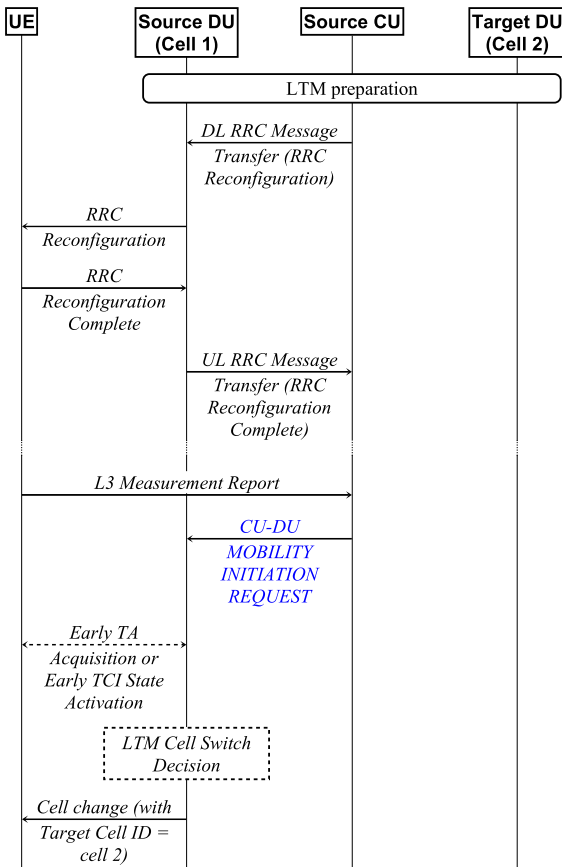


FIGURE 6. LTM procedure based on L3 measurement reporting. Blue font indicates novel behavior specified in Rel. 19.

present the relevant specifications related to L3-based LTM. In a CU/DU split network architecture, L3 measurements are received by the source CU via RRC signaling, while the actual execution of procedures, such as early UL/DL synchronization and cell switching, occurs at L1/L2 of the source DU. This necessitates explicit signaling coordination between the CU and DU. To enable this, a new F1 application protocol (F1AP) message, *CU-DU MOBILITY INITIATION REQUEST*, was introduced [14]. This message allows the CU to instruct the DU to perform LTM-related actions based on L3 measurements. The message includes:

- *Mobility Trigger Information*: It comprises a triggering indication (i.e., which procedure(s) to trigger: early UL/DL synchronization and/or cell switch) and the associated cell and beam information to be used. An overview of the included parameters is summarized in Table 1.
- *Assistance Information*: It provides Layer-3 measurement data reported by the UE for both the serving and candidate cells. This includes a list of SSB measurements, each consisting of an SSB index and corresponding measurement quantity such as RSRP, reference signal received quality (RSRQ), or SINR. For Rel. 19 CSI-RS resource-based measurements,

additional parameters have been introduced to include a list of CSI-RS resource measurements, each consisting of a CSI-RS resource ID and the corresponding measurement quantity.

Both options are supported to offer implementation flexibility across different vendor solutions. When the *Mobility Trigger Information* is included, the CU explicitly specifies which procedure to initiate and with which cell/beam parameters. Alternatively, when *Assistance Information* is provided, the DU may autonomously evaluate the L3 measurements and decide whether and how to perform early synchronization or initiate a cell switch.

The *CU-DU MOBILITY INITIATION REQUEST* thus serves as a key enabler for effective coordination in CU/DU deployments, allowing the DU to execute LTM procedures based on higher-layer decisions or assistive measurements provided by the CU. An example of the LTM procedure based on L3 measurements is illustrated in Fig. 6. In this example, after receiving the L3 measurement report from the UE, the CU sends a *CU-DU LTM MOBILITY INITIATION REQUEST* message to the DU. Based on the information provided in the message, either mobility trigger information or assistance information, the DU can make a decision to initiate early synchronization and/or perform a cell switch. It should be noted that it is possible to have L1 measurements configured for a subset of the candidate cells, e.g., because the UE supports L1 measurements only for specific frequency bands, while relying on L3 measurements for others. It is left to the DU implementation to jointly handle triggers from the CU as well as L1 measurement reports from the UE and make appropriate early synchronization or cell switch decisions.

E. EARLY CSI ACQUISITION

Traditionally, NR handover enhancements have primarily targeted reducing interruption time and increasing reliability. In Rel. 19, a new objective, early CSI acquisition, was introduced to further improve post-handover performance. The aim is to enable immediate high-throughput data transmission and reception following a cell switch by allowing the UE, without delay, to determine appropriate link adaptation parameters, such as, number of layers and precoding weights. This is particularly beneficial in maintaining seamless throughput continuity across cell boundaries. Unlike L1-RSRP measurements, CSI acquisition involves deriving channel characteristics of the target cell, including the CQI, RI, and PMI. These parameters are essential for enabling link adaptation immediately after handover.

Without early CSI acquisition, initial transmissions and receptions after a cell switch rely on conservative fallback configurations, such as transmission without closed loop precoding on large antenna arrays and with low modulation and coding, resulting in reduced throughput until CSI becomes available. Obtaining CSI after the cell switch typically requires the UE to receive CSI-RSs, perform measurements, derive CSI, and report it to the network,

which can introduce delays on the order of hundreds of milliseconds to a few seconds. Such delays can be detrimental to throughput- and latency-sensitive applications, such as extended reality (XR).

To support early CSI acquisition for a candidate cell, one CSI report configuration is provided if the candidate cell supports the early CSI acquisition feature, included in the LTM configuration outside the candidate cell's full RRC configuration. This allows the UE to access CSI configuration information without needing to decode the candidate cell's RRC container. CSI measurements are based on one or two CSI resource sets, associated with the candidate cell, and conveyed via the LTM CSI resource configurations, which are also used for L1-RSRP measurement resources. When only a single resource set is configured, it typically contains non-zero power (NZP) CSI-RS resources used for both channel and interference measurements. If two sets are configured, the first includes NZP CSI-RS resources for channel measurement, while the second includes CSI-IM resources for interference measurement.

Similar to L1-RSRP measurements, both periodic and SP CSI-RS and CSI-IM resources are supported. Similarly, periodic CSI-RS and CSI-IM resources are assumed to be active after the RRC reconfiguration is applied, and for SP CSI-RS and CSI-IM resources, activation is controlled via a MAC CE sent by the source cell, which requires the similar inter-DU/CU coordination, described in Section II-D1 for L1-RSRP measurement, to ensure that the corresponding CSI-RS resource is activated in the candidate cell before the MAC CE transmission. After a cell switch is triggered for a target cell, all active SP CSI-RS and CSI-IM resources for candidate cell(s), except those of the target cell, are deactivated. To deactivate the SP CSI-RS and CSI-IM resources of candidate cell(s) following the UE's successful cell switch, the same inter-DU/CU coordination described in Section II-D2 is reused by the previous serving CU toward the relevant candidate DUs and CUs. For the target cell, the active SP CSI-RS and CSI-IM resources are deactivated after the corresponding CSI reporting. The details of this CSI reporting are provided below.

Only aperiodic CSI reporting is supported for early CSI acquisition. Reporting is implicitly triggered upon reception of the CSC. The UE determines the applicable CSI report configuration based on the *Target Configuration Id* field in the CSC. For baseline UEs, CSI-RS measurement and derivation begin only after CSC reception. Although restricting measurements to the CSI-RS resources associated with the beam indicated in the CSC was discussed during the RAN1 working group discussions, no consensus was reached. Therefore, the UE is allowed to measure all CSI-RS resources within the configured RSs and report the channel resource indicator (CRI) corresponding to the best-measured CSI-RS resource.

For UEs that indicate support for early CSI acquisition prior to receiving the CSC, CSI-RS measurements for one or more candidate cells may begin before CSC reception.

A related discussion considered restricting such measurements to only the activated TCI states prior to cell switch; however, no consensus was reached. As a result, the UE is permitted to measure all configured CSI-RS resources. Nevertheless, the total number of simultaneously active CSI-RS resource ports or resources remains bounded by the UE's legacy capability, requiring the port/resource budget to be shared between the serving cell and all configured candidate cells.

To fulfill the objective of early CSI acquisition, the CSI report is transmitted to the target cell as early as possible, leveraging uplink control information (UCI). The UCI is multiplexed with the earliest available physical uplink shared channel (PUSCH) transmission following the cell switch, depending on the type of the cell switch:

- In RACH-less cell switch, i.e., a cell switch where the random access procedure is skipped, the CSI report is multiplexed with the first PUSCH that carries the RRC Reconfiguration Complete message, either via a configured or dynamic grant.
- In contention-free random access (CFRA)-based cell switch, the report is transmitted on the PUSCH scheduled by the random access response (RAR) or MsgA (in case of 2-step RACH procedure).
- In contention-based random access (CBRA)-based switch, the CSI report is sent on the first PUSCH corresponding to a dynamic grant or a configured grant after the HARQ-ACK transmission corresponding to MsgB (in case of 2-step RACH procedure), or on the first PUSCH scheduled by a physical downlink control channel (PDCCH) transmission addressed to UE's cell radio network temporary identifier (C-RNTI), received in response to message 3 of the RACH procedure. In this case, the PUSCH scheduled by the RAR is not used for CSI reporting, since the message content may be lost due to contention. Therefore, only the PUSCH transmitted after contention resolution was agreed to be used for reliable CSI delivery.

Since the CSI report is transmitted via specific UL transmissions, there may be scenarios in which a valid CSI is not available at the time of reporting, primarily due to insufficient time between CSI-RS resource reception and the reporting slot for the UE to process the measurement and derive the CSI. For baseline UEs, the target cell cannot determine whether CSI acquisition was successful, as the CSC transmission timing is unknown to the target. This behavior differs from legacy CSI acquisition procedures in which the serving cell can know about the CSI timing. To avoid blind UCI decoding at the target cell, when a valid CSI report is not available, the UE transmits a CSI report containing a default indication, the lowest CQI index, which is conventionally used to signal an out-of-range CQI. Since handovers are generally directed toward cells with adequate link quality, such a low CQI value is not expected under normal conditions. Therefore, the use of this default

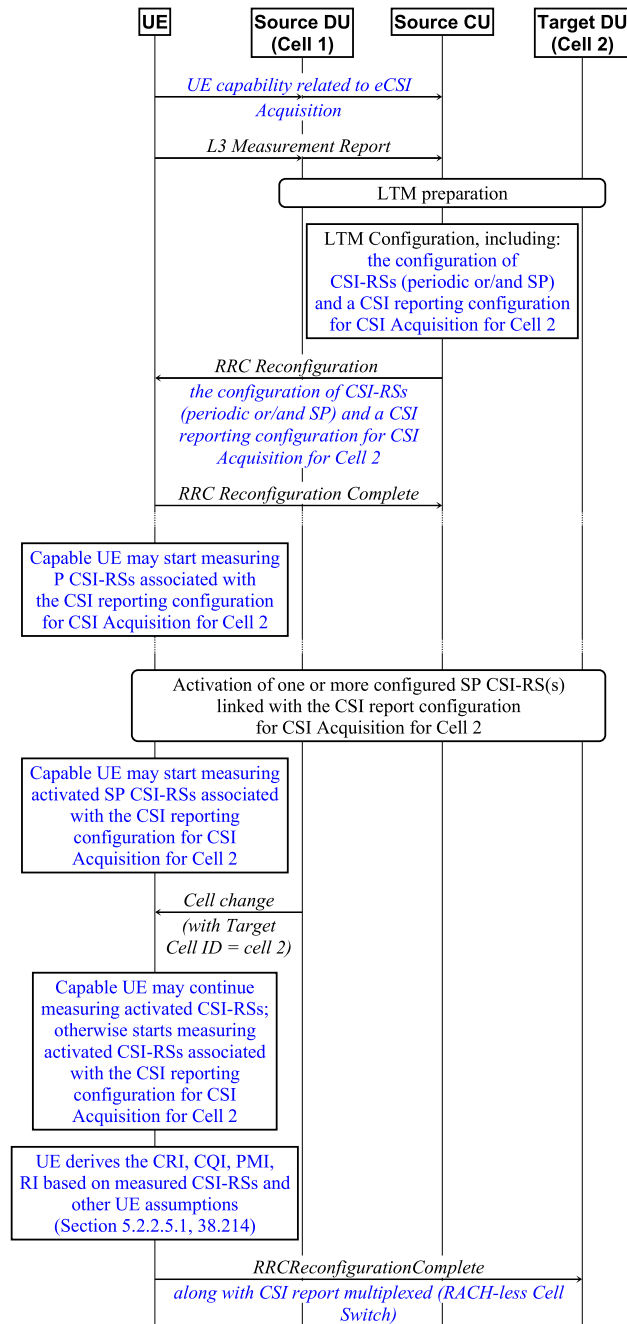


FIGURE 7. An example of early CSI Acquisition procedure for LTM. Blue font indicates novel behavior specified in Rel. 19.

indication was agreed as a reliable method to signal invalid or unavailable CSI in early acquisition scenarios.

Details of early CSI acquisition for LTM can be found in [16]. An example of overall early CSI acquisition showing some of the details discussed above, is shown in Fig. 7 for the RACH-less cell switch with configured grant (CG).

F. CONDITIONAL LTM

Network-triggered LTM, as specified in Rel. 18 and with its Rel. 19 enhancements discussed above, addresses one of

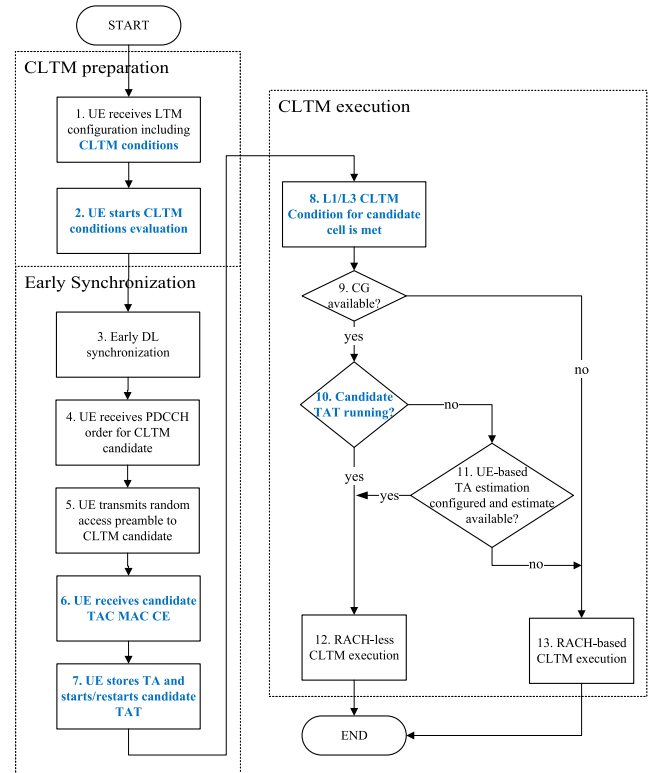


FIGURE 8. UE actions during preparation, early synchronization and execution of CLTM.

the important requirements for mobility mechanisms, namely, low interruption time. On the other hand, its execution requires a successful reception of the CSC by the UE that is triggered by a successful reception of a L3/L1 measurement report by the network. If the radio link quality between the UE and the source cell deteriorates, so that the CSC cannot be delivered to the UE, the cell switch will not be triggered, and the UE will likely experience radio link failure and could potentially be forced to perform reestablishment. This is a well-known concern, not only for LTM, but for all network-triggered mobility procedures. To this end, Rel-16 introduced conditional handover (CHO) [20], where the UE is provided with candidate cell configurations and corresponding conditions for their application. The CHO configuration may be provided to the UE when the link quality to the source cell is still good and the handover may be subsequently triggered autonomously by the UE, if any of the conditions are met.

In Rel. 19, LTM is enhanced with a similar feature, namely Conditional L1/L2-triggered mobility (CLTM), to provide increased mobility robustness. The main principle remains the same as that of the CHO; the UE is configured with and evaluates conditions, which, when met, allow the UE to autonomously trigger a cell switch to the corresponding LTM candidate. Compared to CHO, CLTM has the following distinctive characteristics:

- CLTM is applicable only to intra-CU mobility.

- A CLTM cell switch can be RACH-less, i.e., with reduced interruption time.
- Apart from cell-level L3-RSRP measurements, beam-level L1-RSRP measurements can be used for the cell switch-triggering conditions.
- Subsequent CLTM cell switches are supported without the need to reconfigure the UE.

In Fig. 8, the actions of the UE in different phases of the CLTM procedure are depicted, with the CLTM-specific ones shown with blue font. In the following, a detailed description of the algorithmic procedure is provided, with references to the steps shown in Fig. 8.

During the CLTM preparation phase, in Step 1, the UE receives an LTM configuration which contains one or more CLTM conditions. Since subsequent mobility is supported, i.e., the UE maintains the LTM configuration after a cell switch, the serving cell may change while LTM is configured. Therefore, conditions for a cell switch from the source to any candidate cell and also from any candidate cell to any candidate cell may be provided to the UE. As discussed above, the conditions may be based on L1-RSRP SSB or CSI-RS measurements, configured by the DUs, or L3-RSRP measurements, configured by the CU. Upon reception of the configuration, in Step 2, the UE starts evaluating the conditions.

While the UE is evaluating CLTM conditions, it can receive a command from the network to perform early DL synchronization (Step 3) via a candidate TCI state activation command MAC CE and/or a command to perform early UL synchronization via a PDCCH order, as shown in Step 4 (more details on early synchronization procedures can be found in [1]). Early DL synchronization works as in LTM, with some additional implications on condition evaluation. More specifically, in FR2, including frequency bands from 24.25 GHz to 71 GHz, after TCI state activation for at least one candidate, the UE is not required to measure RSs from candidate cells without active TCI states and, therefore, does not evaluate CLTM conditions for switching to them. Hence, the network can use TCI state activation to dynamically control CLTM condition evaluation.

The impact of potentially UE-triggered cell switching is more prominent in early UL synchronization. In non-conditional LTM, the TA value obtained during early UL synchronization, based on the transmitted random access preamble by the UE (Step 5), is provided by the source cell to the UE at the time of the cell switch within the CSC. On the contrary, for CLTM, the source cell must provide the available TA to the UE as soon as possible, so that it is available for a potential CLTM cell switch, which can occur anytime, based on a configured CLTM condition being met. To this end, a new candidate timing advance command (TAC) MAC CE was introduced in Rel. 19, sent from the serving cell to the UE and carrying the TA and the ID of the candidate cell to which it corresponds. Upon reception of the candidate TAC MAC CE (Step 6), in Step 7, the UE stores the TA value and starts a

corresponding time alignment timer (TAT) which determines the duration for which the TA value remains valid.

When one or more CLTM conditions are met (Step 8), the CLTM execution procedure is triggered. To determine whether the cell switch will be RACH-less or RACH-based, the UE checks first, in Step 9, whether it has a valid grant for the initial transmission to the target cell (transmission of the *RRCReconfigurationComplete* message). We note that only CG-based RACH-less CLTM is supported, as the timely provision of a dynamic grant (DG) is challenging due to the unknown timing of the cell switch on the network side and requires further optimization. If the triggered condition is L1-RSRP-based, the UE checks if it has a valid CG available for any of the RSs associated with a triggered condition. If the triggered condition is L3-RSRP-based, the UE checks whether it has a CG available for any SSB whose L1-RSRP is above a configured threshold. If a valid grant is available, the UE subsequently checks whether a valid TA is available for the triggered candidate cell, whereas if not, a RACH-based cell switch is triggered (Step 13).

For determining the availability of a valid TA, in Step 10, the UE checks if the candidate TAT of the triggered cell is running in the first available CG occasion, which implies that a candidate TAC MAC CE for the triggered candidate cell was previously received and the TAT does not expire before the first uplink transmission. If so, the UE uses the corresponding TA to perform RACH-less cell switch (Step 12); otherwise, the UE may use a UE-based TA estimate (see [1]), if available (Step 11). If none of the above holds, the UE falls back to RACH-based cell switch (Step 13).

Finally, it should be mentioned that, similar to legacy L3 measurement-based mobility, the network can trigger a cell switch at any point of the described procedure for any of the involved cells by sending a CSC to the UE. Therefore, it should be noted that any CLTM candidate cell is also a conventional LTM candidate.

III. REL. 20 AND 6G TOPICS

A. LTM WITH CA

1) BACKGROUND ON CA

CA is one of the most successful features of 5G that is deployed widely. CA enables improved throughput for the UE through aggregating carriers; as such, the bandwidth that the UE can be scheduled on is increased. The cells that are added to the UE configuration under CA procedure are called secondary cells (SCells). UE may be configured with multiple SCells. SCells with PCells constitute a cell group. The configuration of SCells mainly differs in lower layers, such as the physical layer.

A typical deployment for an SCell would be to deploy them on the same master as the PCell, called a co-located deployment. This would mean that radio characteristics, for example, the timing advance or signal strength, would be quite similar to the PCell. In case SCell is used for DL only where DL ACK is sent over PCell, the timing advance might not be necessary.

For CA configuration UE receives the SCell configuration in a de-activated way. The network can then activate the SCell configuration through an SCell activation command. During the handover procedure, UE can be configured with deactivated SCell or with a Rel. 16 feature to directly activate SCells. This would mean that, after handover completion, UE is required to measure and activate the SCell. In case the timing advance information is necessary, the network will configure the SCell in a deactivated way and acquire the TA for the UE before activating the SCell. The delay in SCell activation may result in a throughput dip right after handover. Rel. 20 aims to improve this through extending the reduced interruption time of LTM to also to LTM cell switch with CA.

2) ENHANCEMENTS FOR RELEASE 20

Rel. 18 and Rel. 19 enhancements are mainly intended for PCell and PSCell. The enhancements have overlooked SCells. Current Rel. 20 WI objective [21] captures that the network can configure the UE with measurement reports specific to candidate SCells. The network, using the measurements of candidate SCells and the user plane activity at the UE side, may indicate to the UE which SCells shall be directly activated, along with the TCI states to be used for each SCell.

Current work item description does not cover the non-co-located case to also consider early TA acquisition procedure for candidate SCells. We foresee that for the complete applicability of LTM with CA solution network-wide, the early TA acquisition procedure shall be integrated.

B. DYNAMIC MEASUREMENT REPORT/RESOURCE UPDATE

In both Rel. 18 and Rel. 19 LTM, the L1 measurement and reporting procedures rely on common CSI resource configurations, i.e., *LTM-CSI-ResourceConfig* as defined in [19]. Each resource configuration contains a resource set comprising a list of RS indices, which may include SSB resource indices or CSI-RS resource indices. These resource sets are referenced by report configurations that can be configured for periodic, semi-persistent, aperiodic, or event-triggered reporting, supporting L1-RSRP measurements or early CSI acquisition, as detailed in Sections II-D1, and II-E.

Such common resource configurations are constructed by the source gNB (or CU) through coordination with all candidate cells prepared for the UE, including cells from both the same and inter-CU gNBs. These configurations are intended to be shared among candidate cells to facilitate seamless cell transitions without requiring dedicated RRC reconfiguration messages, also known as subsequent LTM cell switches. In other words, when the UE moves to a candidate cell, the reporting configurations needed for the next cell switch refer to the same common resource configurations. If any candidate LTM cell requires a change in the configuration, coordination among all participating network nodes is necessary. The serving cell then prepares

the updated configuration based on the coordination outcome and sends it to the UE.

This coordination process can introduce drawbacks in LTM, particularly because mobility here involves beam management and measurements across multiple candidate cells. Due to the nature of beam-based mobility, the set of beams or cells to be measured and reported may vary for subsequent cell switches. Without dynamic selection, configuring the UE to measure all beams from all candidate cells always would incur significant and unnecessary measurement overhead at the UE side. For periodic reporting, this also results in considerable reporting resource overhead. Typically, due to UE limitations on the number of beams it can measure and report, the network restricts the number of reporting configurations and L1 measurement resources accordingly. As the UE moves or channel conditions change, updating these measurements and reporting configurations becomes necessary. Even for UEs capable of measuring a large number of beams, preconfiguring all potential measurement resources increases power consumption significantly.

Currently, any changes to measurement resource configurations require inter-node (between CUs and DUs) signaling and corresponding serving cell actions to provide updated RRC configurations, which limits the efficiency of LTM. Therefore, in Rel. 20, a new objective has been agreed upon [21] to enable a faster update mechanism for measurement and reporting configurations using L2 control signaling, i.e., MAC CE. This approach allows selecting subsets of measurement resources, such as specific beams from certain candidate cells, for each reporting configuration. In other words, it could enable activating or deactivating different sets of measurements and corresponding reporting dynamically, without the need for RRC reconfiguration. This enhancement is expected to reduce signaling overhead and improve the efficiency of LTM-related measurements and reporting.

C. 6G MOBILITY

The 6G study started in October 2025 within the 3GPP RAN2 working group, where mobility aspects are typically handled. The 6G mobility framework is envisioned as an evolution of previous generations, aiming towards a unified, efficient, and intelligent mobility management system that is robust and capable of ensuring low interruption time with service continuity. This effort builds on the strengths of earlier generations while addressing their limitations through carefully selected enhancements rather than incremental feature accumulation.

So far, the main focus of the discussions has been on reviewing the wide range of mobility enhancements introduced across different releases of 5G and determining which of these should be down selected for 6G. These enhancements include, but are not limited to, CHO, dual active protocol stack (DAPS), LTM enhancements such as early synchronization procedures, L1 measurement enhancements for mobility, and subsequent mobility mechanisms.

The overarching objective is to achieve improved handover performance while keeping network complexity at a manageable level. Due to the additional complexity and limited benefit to cost trade off, DAPS has been excluded from further consideration in the 6G mobility study. As a result, the discussions have converged on a smaller set of key topics, including mobility with pre-configuration, handover signaling design, and the unification of measurements for mobility.

At the same time, the 6G mobility design explicitly aims to address the shortcomings identified in 5G mobility. In particular, the following issues have been agreed upon in recent discussions: mobility procedures introduced across different releases have not always been well integrated with each other; different protocol layers such as RRC, MAC, and L1 have overlapping responsibilities; certain mobility related configurations are repeated; signaling overhead has increased due to some procedures, such as subsequent mobility introduced for LTM; the scope of resilience remains limited; and pre-configuration based mobility mechanisms introduce non negligible resource reservation at the network side. These observations highlight the need for a more holistic and coordinated mobility framework in 6G, where functionality is better aligned across layers and procedures are designed with both performance and efficiency in mind.

In this context, handover with pre-configuration (e.g. CHO or LTM, known from 5G) is being carefully reconsidered for 6G. Pre-configuration can provide robustness benefits by allowing the UE to be prepared with candidate cell configurations in advance and to use them instantaneously when the source cell's radio link quality degrades. However, these benefits come at the cost of increased radio resource reservation in the network. Consequently, pre-configuration based mobility must be studied with particular attention to its impact on both the UE and the network, and potential refinements are required to make it suitable for adoption in 6G.

To this extent, the design of pre-configuration itself is a central topic in the ongoing discussions. One consideration is whether pre-configuration should continue to rely on storing the target cell *RRCReconfiguration* message as an OCTET string, or whether enhancements such as modularized RRC or delta signaling mechanisms should be introduced. Such enhancements could allow partial pre-configuration, thereby limiting unnecessary resource reservation. Moreover, improvements in RRC design related to *RRCReconfiguration* can help reduce the handover command size, which has a direct impact on handover robustness. A more refined RRC design may also enable the specification of more granular processing requirements, contributing to reduced handover interruption time and improved service continuity.

Another complementary direction for reducing interruption time, particularly in the first release of 6G, is the support of early uplink and downlink synchronization procedures. These mechanisms enable the UE to become "radio" ready with respect to the target cell before receiving the handover

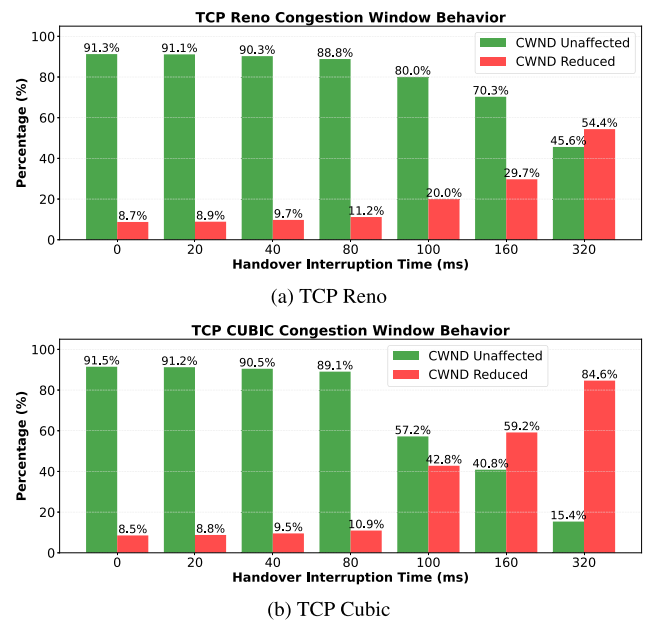


FIGURE 9. Change in congestion window size during handovers for different handover interruption time durations. Red bars indicate the percentage of handovers where the TCP operation is impacted through RTO expiry or duplicate ACKs. Green bars show the percentage of handovers where this is not the case.

command, thereby shortening the interruption phase during mobility execution.

More broadly, a fundamental challenge in designing 6G mobility lies in the fact that use cases requiring enhanced mobility performance were not widely deployed in previous generations. Therefore, the 6G mobility framework should primarily address current and well understood mobility challenges, while at the same time being sufficiently flexible to accommodate future requirements. The applicability of ongoing Release 20 work on LTM to 6G mobility remains to be seen; nevertheless, agreements reached in Release 20 discussions, such as those related to dynamic measurement configurations, may serve as valuable input to ensure that relevant enhancements are incorporated into 6G from the outset.

IV. SIMULATION ANALYSIS OF IMPACT ON TCP DUE TO INTERRUPTION TIME

In LTM and in other types of mobility procedures in general, the goal is to ensure seamless connectivity from the perspective of the UEs. However, during baseline handover (BHO), the UE experiences a short drop in connection for around 80-100 ms [22], which is the interruption time. During this time, the UE is neither connected to the source cell nor to the target cell and therefore, there is no user plane data transmission. LTM reduces the interruption time from 80-100 ms to around 20 ms [18] due to early preparation of the target cell(s) and due to LTM execution command being a MAC CE instead of an RRC message. However, the interruption time can be even longer due to wrong handover

parameter settings or due to long recovery procedures in case of a mobility failure.

In this section, we discuss the effect of such different interruption time durations on TCP Reno and TCP CUBIC. Interruption time can affect TCP performance mainly by causing the TCP RTO timer expiry. After transmission of every packet, TCP starts an RTO timer and waits for an acknowledgement (ACK) reception throughout the timer duration. If there isn't any ACK reception upon the timer expiry, then TCP interprets it as a congestion along the network route, and reduces the congestion window (cwnd) size to the smallest possible value, which results in a severe degradation in throughput. In Linux kernel, the default minimum value of the RTO timer is 200 ms [9]. It can also be configured to lower values, such as to 100 ms, or even to 50 ms [23]. In case of the low RTO values, the handover interruption time becomes critical. If the interruption time is longer than the RTO at the time of the handover, then whenever there is a handover, there is also likely an accompanying RTO expiry, which results in suboptimal throughput. In this case, switching from BHO to LTM can bring a significant improvement to TCP performance since the interruption time in LTM is much smaller when compared to BHO (20 ms versus 100 ms). Therefore, RTO expiry event becomes much more unlikely in the case of LTM.

In Section IV-A, we analyze the results of the system-level simulations. In Appendix A, we give some background information on TCP Reno and CUBIC, and provide a more detailed description of the simulation setup.

A. RESULTS AND DISCUSSIONS

In light of the background information in Appendix A-A, we grouped the congestion window behavior into two categories in Fig. 9. The green columns, *cwnd Unaffected*, show the percentage of the times that the TCP is in slow start or congestion avoidance during handovers and maintains this behavior after the handover as well. The red columns, *cwnd Reduced*, on the other hand, show the amount of times that the RTO expired or TCP was in loss recovery during handovers. While constructing the bar plots, we evaluated how the handovers affect the cwnd behavior. To that end, we collected the TCP traces and the handover information of all the UEs in our simulation campaign. Then, we overlayed the handover information onto the TCP traces. Afterwards, we determined the last cwnd size of the UE right before the beginning of the handover and the first cwnd size right after the end of the handover. Finally, we evaluated the change in those two cwnd samples and obtained Fig. 9. It should also be noted that the different handover interruption times in Fig. 9 correspond to different mobility optimization approaches, where 0 ms means ideal cell switch (or bandwidth switch), 20 ms is the LTM interruption with all the early sync procedures, 40 ms corresponds to LTM with all the early sync procedures except the early UL sync, 80 ms and 100 ms are the typical interruption time durations for BHO,

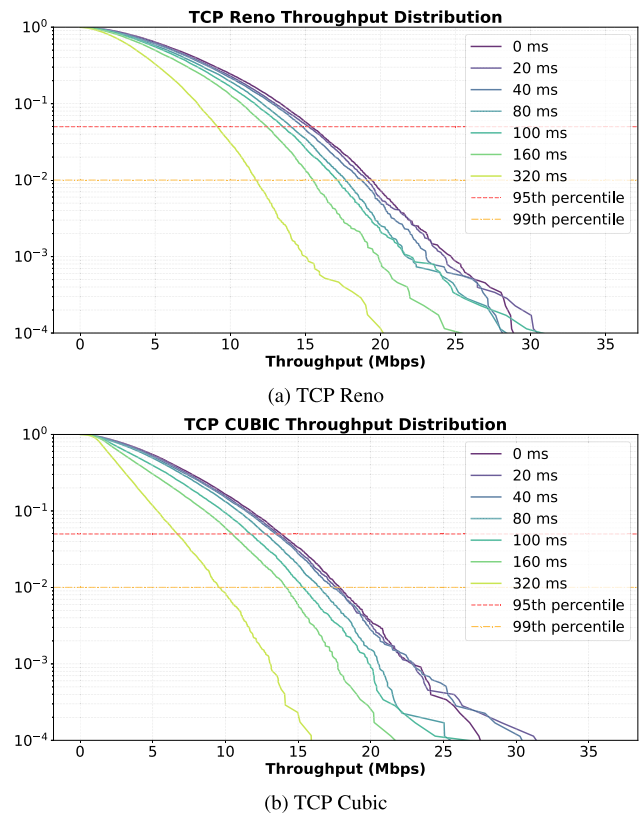


FIGURE 10. Throughput during handovers for different handover interruption time durations.

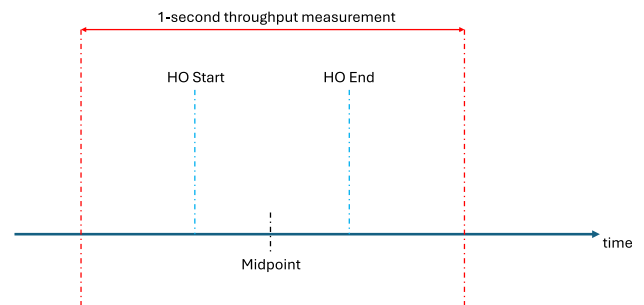


FIGURE 11. 1-second throughput calculation window. HO Start corresponds to the starting of the handover. HO End corresponds to the end of the handover. Midpoint corresponds to the midpoint of the handover interruption time and the 1-second throughput window. The midpoints of the throughput window and handover interruption time are aligned.

160 ms corresponds to BHO with contention-based random access procedure with many retransmissions and 320 ms corresponds to the re-establishment procedure delay. These time duration explanations are the same for all the figures presented in this section, namely for Fig. 9, Fig. 10, Fig. 12 and Fig. 13.

In Fig. 9, we can observe that the cwnd percentages stay the same for both Reno and CUBIC for handover interruption time values from 0 ms to 80 ms. However, when transitioning from 80 ms to 100 ms, we observe that *cwnd Reduced* rate

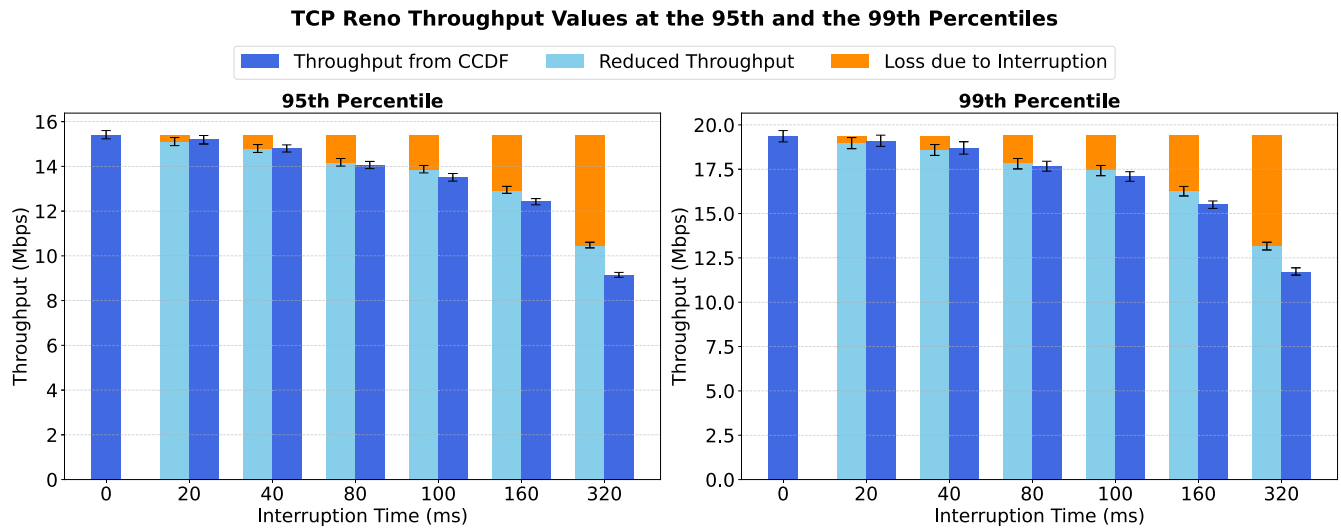


FIGURE 12. TCP Reno throughput distribution at higher percentiles. The error bars correspond to the 95% confidence intervals.

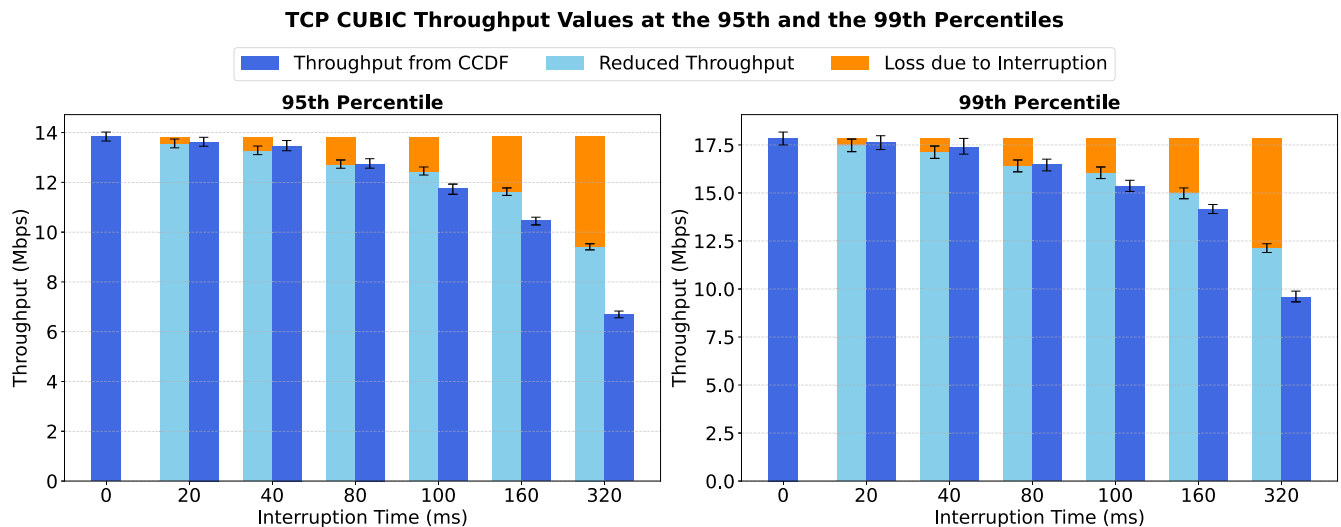


FIGURE 13. TCP CUBIC throughput distribution at higher percentiles. The error bars correspond to the 95% confidence intervals.

almost doubles for TCP Reno (Fig. 9a) by going from 11.2 % to 20.0 % and almost quadruples for TCP CUBIC (Fig. 9b) by going from 10.9 % to 42.8 %. Similarly, we observe a corresponding drop in *cwnd Unaffected* percentages both in TCP Reno and CUBIC, from 88.8 % to 80.0 % in case of TCP Reno and from 89.1 % to 57.2 % in case of TCP CUBIC. From 100 ms interruption time onward, we observe that congestion window size is more and more negatively impacted by the increasing handover interruption time duration, with TCP CUBIC being the one that is more sensitive to handover interruption time. Finally, the reason we have the breaking point at 100 ms is because the initial TCP RTO value is set to 100 ms. Therefore, once the handover interruption time reaches to 100 ms,

it becomes much more likely for RTO to expire, causing *cwnd* to collapse. Increasing the handover interruption time further increases the probability of RTO expiry, which further increases the *cwnd Reduced* percentage.

Investigating the CWND only shows that the interruption time has an impact on TCP operation. More practically, change in CWND would mean change in throughput. In Fig. 10, it is possible to observe the throughput distribution at the time of handover for different interruption time durations for both TCP Reno and CUBIC. The way the throughput is calculated is visualized in Fig. 11. In short, the transmitted data is measured in a 1-second interval and the handover is placed in the middle of that interval. This is carried out for all the handovers and the resulting complementary

cumulative distribution functions (CCDF) are constructed. We observed significantly different throughput values for different handover interruption time durations at the higher percentiles of the CCDF. We plotted two such percentiles in Fig. 10, the 95th, and the 99th percentiles. For ease of inspection, the throughput values at those percentiles are provided in Fig. 12 for TCP Reno and in Fig. 13 for TCP CUBIC.

In addition to providing the throughput at a given percentile (*Throughput from CCDF*), Figures 12 and 13 also provide the change in measured throughput in simulations with respect to the baseline throughput at 0 ms interruption time. To distinguish throughput reduction due to interruption time and TCP behavior (such as reduced cwnd), we added a stacked bar next to each throughput bar. The stacked bar has an upper and lower part, namely *Loss due to Interruption* and *Reduced Throughput* respectively. The upper part, *Loss due to Interruption*, represents the percentage of handover interruption compared to 1 s interval in terms of throughput. This is subtracted from the 0 ms *Throughput from CCDF* to form the lower part. The lower part, *Reduced Throughput*, represents the throughput of 0 ms scenario with percentage of throughput reduced due to handover interruption time. Through this, the lower part can be compared with the *Throughput from CCDF* for each interruption time, to observe the throughput degradation purely due to non-interruption related aspects, which is hypothesized to be representing the impact on TCP behavior. In other words, one should compare light and dark blue bars to infer the throughput degradation due to TCP being impacted by the handover. From this regard Figures 12 and 13 are aligned with Fig. 9 as we explain in the next paragraph.

Both in Fig. 12 and in Fig. 13, we observe that for interruption time values up to 80 ms, *Reduced Throughput* and the *Throughput from CCDF* have overlapping confidence intervals for both percentiles. This means that the reduction in throughput for interruption time values 0, 20, 40 and 80 ms stems from the interruption time itself and other TCP related decrease is not statistically significant to be detected. For example, when the interruption time is equal to 80 ms, it would be reasonable to expect around 8% loss in throughput since there is no user plane data transmission for 80 ms in the 1-second throughput window. This is exactly the case for interruption times lower than 100 ms. Hence, our first observation on the impact of mobility on TCP performance is as follows: *Throughput reduces as the interruption time increases since the user plane connection between the UE and the gNB is cut off for a longer time.*

On the other hand, for interruption time values 100 ms and higher, we observe that the throughput values taken from the CCDFs are lower than their corresponding *Reduced Throughput* values with no overlap in their confidence intervals, except for 100 ms in 99th percentile in Fig. 12. In fact, the difference between the *Reduced Throughput* and the *Throughput from CCDF* grows larger as the interruption time increases. The reason of this difference can

be understood by looking at Fig. 9. Here, we observe that *CWND Reduced* starts to increase drastically for both TCP Reno and CUBIC starting from 100 ms interruption time, since the initial TCP RTO value is set to 100 ms in our experiments. Therefore, we make another observation on the effect of mobility on TCP: *If the interruption time is the same or larger than the initial TCP RTO value, then throughput suffers because cwnd size reduction becomes much more likely.*

Figures 12 and 13 give us insight about the impact of LTM and BHO on TCP as well. We see that the *Throughput from CCDF* value at 20 ms interruption time has an overlapping confidence interval with the throughput at 0 ms, from which we conclude that there is not a meaningful statistical difference between the performance of LTM and ideal handover with no interruption. However, this is not the case for typical BHO interruption time values of 80 and 100 ms. In these interruption times, we observe a throughput degradation with respect to 0 ms interruption time. Hence, we make our final observation: *LTM improves TCP throughput compared to BHO by reducing the interruption time.*

Of course, this simulation campaign is carried out and the results are discussed in FR1 band. Therefore, one would wonder whether the same observations would be made in FR2 band as well. However, it should be noted that the focus of this study has been on the interruption time. Although the underlying channel is different in FR2 and there would be narrower beams, from the perspective of this study, what matters is the interruption time. If we assume that the coverage is enough, then the differences wouldn't matter much. Noting that we work under full buffer traffic, what we would observe in FR2 would be the scaling of throughput. Hence, it would be reasonable to expect a similar behavior in FR2 band as well. However, investigating the mobility impact on TCP in FR2 is an interesting aspect for future work.

Another aspect for future work is the analysis of mobility impact on the other TCP performance metrics. During a handover, MAC and RLC layers are flushed. Depending on whether the RLC layer is operating in AM or in unacknowledged mode (UM), packets may be lost during a handover. Specifically, if the RLC layer is operating in UM, then the packets in the RLC buffer are lost because the packets are not tracked. However, if it is in AM, then after flushing its buffers, RLC layer can take the lost packets from PDCP layer and transmit them again because PDCP layer holds the packets in its buffer until it receives a successful transmission indication from the RLC layer in case of AM operation. Therefore, as can be seen, depending on the protocol stack configuration, there can be packet loss during handovers, which would cause retransmissions in the TCP layer and that would mean the TCP congestion window size would be reduced. Furthermore, latency would increase because of packet loss or retransmissions caused by handovers. Therefore, investigating mobility induced packet loss and latency in the TCP layer is an interesting aspect that we will pursue in future work.

V. CONCLUSION

This paper has provided a comprehensive analysis of the significant enhancements made to LTM in 3GPP Rel. 19, demonstrating the evolution from the intra-CU baseline solution of Rel. 18 to a more robust and versatile mobility framework. The Rel. 19 enhancements successfully address critical limitations through four main advancement areas: deployment scenario expansion via inter-CU LTM, resource efficiency improvements through event-based reporting, enhanced mobility robustness with CSI-RS measurements and conditional LTM, and cell-edge throughput optimization via early CSI acquisition. The inter-CU LTM implementation represents a major step forward in practical deployment applicability, enabling seamless mobility across different CUs while maintaining the low interruption time benefits of LTM. The sophisticated security key management and L2 reset mechanisms ensure secure and reliable operation across CU boundaries, while the enhanced signaling procedures over Xn interfaces provide the necessary coordination for successful inter-CU mobility. Our simulation analysis reveals the substantial impact of interruption time reduction on transport layer performance, with LTM's ~ 20 ms interruption time providing significant TCP throughput improvements compared to baseline handover's ~ 100 ms interruption. The results demonstrate that when TCP RTO timers are configured below 100 ms, the probability of RTO expiry during handover becomes critical, making LTM's reduced interruption time particularly valuable for maintaining optimal transport layer performance. Looking toward the future, Rel. 20's focus on LTM with CA and dynamic measurement configuration represents the next logical evolution, addressing cell edge throughput and resource overhead challenges. The preliminary 6G mobility framework discussions suggest a continued emphasis on unified, intelligent mobility management that builds upon the LTM foundation while addressing emerging requirements for ultra-reliable, low-latency communications. The comprehensive enhancements introduced in Rel. 19 position LTM as a mature and practical mobility solution that significantly advances the state-of-the-art in cellular mobility management, providing a solid foundation for future 5G-Advanced and 6G mobility innovations.

APPENDIX A SYSTEMLEVEL SIMULATION DISCUSSIONS

This section gives more details regarding the simulations in Section IV. In Appendix A-A, we provide some background information on TCP Reno and CUBIC and in Appendix A-B, we provide details on our system-level simulations setup.

A. BACKGROUND ON TCP RENO AND TP CUBIC

TCP regulates the amount of data that the sender is allowed to send by maintaining a congestion window. Since we model UEs that use full buffer traffic, they always try to utilize the entire congestion window. Therefore, the size and the behavior of the congestion window directly impacts the

amount of data that a UE can transmit. Both TCP Reno and CUBIC have the following common properties in their congestion control behavior:

- *Slow start*: Slow start is characterized by the rapid increase in the congestion window size up until to a threshold (i.e. slow start threshold). Congestion window size is doubled at the end of every RTT. In our implementation, both TCP CUBIC and Reno use the Reno-style slow start mechanism.
- *Congestion avoidance*: After the slow start follows the congestion avoidance phase. During this phase, Reno uses a linear increase function, while Cubic uses a cubic function to adjust the congestion window size. For details of these congestion window increase functions, the reader is encouraged to read further [8], [24].
- *Loss recovery based on 3-duplicate ACKs*: Whenever it receives a packet, TCP receiver transmits an ACK message back to the sender to indicate bytes up until to a certain point have been successfully received. However, if there is a packet loss in the middle of a transmission, then the TCP receiver acknowledges only the last successfully received packet, thus never acknowledging the lost packet in between and the other successfully received packets after that. This results in the TCP receiver sending the same ACK message multiple times back to back. Upon receiving four such ACK messages (which means three duplicate ACKs reception, the first ACK is the original and three duplicates follow after it), TCP sender infers there is a packet loss in the network and immediately retransmits the missing packet instead of waiting for the RTO timer to expire. After that, TCP Reno reduces the congestion window size to approximately half of its previous size whereas CUBIC follows a less aggressive route, reducing it by a factor of β , where it is typically set to 0.7 or to 0.8.
- *RTO expiry*: Both TCP CUBIC and Reno employ an RTO timer to keep track of the received ACKs. After receiving every ACK, the RTO timer is reset to its calculated value and if there isn't any ACK reception until the timer expires, then the congestion window collapses, meaning that it is set to 1 segment size.

B. SYSTEM MODEL

In order to analyze the effect of interruption time on TCP performance, we have conducted simulation studies using an in-house 3GPP-compliant system-level simulator [25]. The system-level simulator can simulate a wide range of scenarios such as high-speed train [26], [3], highway [2] and Manhattan grid [27] in different frequency bands. Furthermore, the cellular protocol stack, the channel model and the mobility procedures are also in line with the 3GPP specifications.

Using this system-level simulator, we have simulated a bulk TCP data transfer in DL direction for each UE with

21 UEs traveling with a constant speed in a random direction. Once their random travel direction is determined in the beginning of the simulation, the UEs always preserve their direction. To prevent the UEs from bouncing back once they reach to the edges of the simulation map, we implemented a wrap-around feature [2], which means that the simulation map is translated to left, right, up and down so that when a UE exits the map from one side enters back from the opposite side. The same thing happens for the signal propagation as well. Therefore, there is interference also at the map edges. Finally, during handovers, the UE's experience different outage periods, which correspond to different handover interruption times.

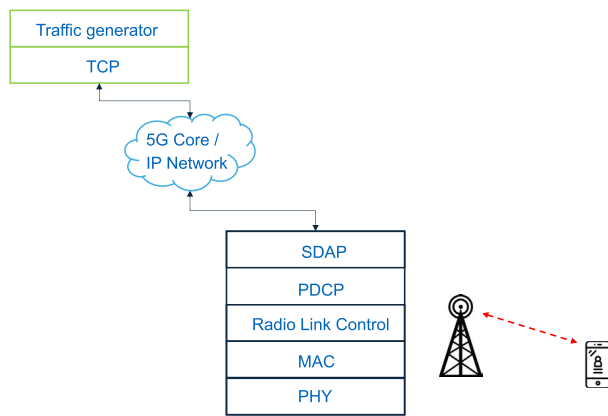


FIGURE 14. The DL protocol stack of the data flow.

Fig. 14 shows the entire protocol stack of a UE inside the simulator. In Fig. 14, the traffic generator corresponds to the application that generates data. We used full buffer traffic model [3] so that there is always data to transmit. In the TCP layer, we tested two different TCP variants, namely TCP CUBIC [8] and TCP Reno [24]. Once the packet is generated by a server, it first traverses the IP network and then it enters to the 5G Core, after which it arrives to the SDAP layer of the serving cell. In our implementation, since our focus is on the radio part modeling, we modeled the core network/IP network delay as a constant 15 ms delay based on the measurements provided in [28]. At the RAN protocol stack, we model the retransmissions at the MAC layer and RLC layer is in UM, therefore a TCP layer retransmission is triggered only when all of the retransmissions at the MAC layer fail. Table 2 lists rest of the important simulation parameters.

The network is arranged in a 7-site hexagonal grid layout [29] with inter-site distance (ISD) set to 200 meters [30]. The channel model is 3GPP-compliant and more details regarding it can be found in [31]. Apart from that, the handover parameter configuration (A3 Event offset, A3 Event hysteresis, Time-to-Trigger) can be treated as the default configuration across other system-level simulations [32].

TABLE 2. List of the important simulation parameters used for analyzing the effect of handover interruption on TCP performance.

Parameter	Value
Channel model	TR38.901 NLOS
Carrier Frequency	4 GHz
Duplexing	FDD
Bandwidth	10 MHz
Subcarrier Spacing	15 kHz
Number of UE's	21
UE Speed	90 km/h
A3 Event Offset	3 dB
Time-to-Trigger	160 ms
Network Layout	7-site hexagonal grid
Inter-site Distance	200 meters
TCP RTO	100 ms
Packet Size	1.5 kB
Simulation duration	25 seconds
Random realizations	100

TCP RTO value selection is based on [23]. Lastly, the duration of a single simulation is set to 25 seconds.

Simulation campaigns are carried out for different handover interruption times to investigate its effect on TCP performance. The value is decreased from 100 to 20 ms to do a sweep analysis between baseline and LTM interruption times. Values 160 and 320 ms are added to show the impact of wrong handover settings or for recovery procedures. Finally, 0 ms interruption is provided to benchmark the effect of different interruption times against an ideal handover that has no interruption. For each case, 100 simulations with different random seeds are run and then then for each handover interruption time duration, approximately 17,000 handover events are analyzed across those 100 simulations. Using these samples, the results in Section IV-A are presented.

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REFERENCES

- [1] B. Khodapanah, S. Goyal, M. Gursu, J. Stanczak, A. Kakkavas, R. Temelli, A. Badalioglu, P. Spapis, and C. Majumdar, "On L1/L2-triggered mobility in 3GPP release 18 and beyond," *IEEE Access*, vol. 12, pp. 171790–171806, 2024.
- [2] D. Atik, M. Gursu, B. Khodapanah, F. Mehmeti, and W. Kellerer, "On the impact of handovers on packet reliability in mobile networks," in *Proc. IEEE Int. Conf. Commun.*, Jun. 2025, pp. 1–6.
- [3] D. Atik, M. Gursu, F. Mehmeti, B. Khodapanah, and W. Kellerer, "Analysis of the rural network deployment to achieve end-to-end latency requirements of future railway mobile communication systems," *Comput. Netw.*, vol. 272, Nov. 2025, Art. no. 111649.
- [4] J. Wang, Y. Zheng, Y. Ni, C. Xu, F. Qian, W. Li, W. Jiang, Y. Cheng, Z. Cheng, Y. Li, X. Xie, Y. Sun, and Z. Wang, "An active-passive measurement study of TCP performance over LTE on high-speed rails," in *Proc. 25th Annu. Int. Conf. Mobile Comput. Netw.*, Aug. 2019, pp. 1–16.
- [5] Q. Liu, K. Xu, H. Wang, M. Shen, L. Li, and Q. Xiao, "Measurement, modeling, and analysis of TCP in high-speed mobility scenarios," in *Proc. IEEE 36th Int. Conf. Distrib. Comput. Syst. (ICDCS)*, Jun. 2016, pp. 629–638.
- [6] K. Winstein, A. Sivaraman, and H. Balakrishnan, "Stochastic forecasts achieve high throughput and low delay over cellular networks," in *Proc. 10th USENIX Symp. Networked Syst. Design Implement. (NSDI)*, 2013, pp. 459–472.

- [7] Y. Zaki, T. Pötsch, J. Chen, L. Subramanian, and C. Görg, "Adaptive congestion control for unpredictable cellular networks," in *Proc. ACM Conf. Special Interest Group Data Commun.*, Aug. 2015, pp. 509–522.
- [8] S. Ha, I. Rhee, and L. Xu, "CUBIC: A new TCP-friendly high-speed TCP variant," *ACM SIGOPS Operating Syst. Rev.*, vol. 42, no. 5, pp. 64–74, Jul. 2008.
- [9] Z. Liu, Q. Deng, Z. Tan, Z. Qian, X. Zhang, A. Swami, and S. V. Krishnamurthy, "M2HO: Mitigating the adverse effects of 5G handovers on TCP," in *Proc. 30th Annu. Int. Conf. Mobile Comput. Netw.*, Dec. 2024, pp. 1089–1103.
- [10] *Revised Work Item: NR Mobility Enhancements Phase 4*, Standard WID RP-252111, 3GPP, Sep. 2025.
- [11] *NR; NR and NG-RAN Overall description; Stage-2*, Standard TS 38.300, v19.1.0, Jan. 2026.
- [12] *NG-RAN; Xn Application Protocol (XnAp)*, Standard TS 38.423, v19.1.0, Dec. 2025.
- [13] *Security Architecture and Procedures for 5G System*, Standard TS 33.501, v20.0.0, 3GPP, Jan. 2026.
- [14] *NG-RAN; F1 Application Protocol (F1AP)*, Standard TS 38.473, v19.1.0, 3GPP, Dec. 2025.
- [15] *NR; Medium Access Control (MAC) Protocol Specification*, Standard TS 38.321, v19.1.0, 3GPP, Jan. 2026.
- [16] *NR; Physical Layer Procedures for Data*, Standard TS 38.214, v19.2.0, 3GPP, Jan. 2026.
- [17] *NR; Multiplexing and Channel Coding*, Standard TS 38.212, v19.2.0, 3GPP, Jan. 2026.
- [18] *NR; Requirements for Support of Radio Resource Management*, Standard TS 38.133, v19.3.0, 3GPP, Jan. 2026.
- [19] *NR; Radio Resource Control (RRC); Protocol Specification*, Standard TS 38.331, v19.1.0, 3GPP, Jan. 2026.
- [20] J. Stanczak, U. Karabulut, and A. Awada, "Conditional handover in 5G—principles, future use cases and FR2 performance," in *Proc. Int. Wireless Commun. Mobile Comput. (IWCMC)*, May 2022, pp. 660–665.
- [21] *New WI: NR Mobility Enhancements Phase 5*, Standard WID RP-251865, 3GPP, Jun. 2025.
- [22] *Moderator Summary for Multi-Beam Enhancement: EVM*, document TSG-RAN WG1 Meeting 102, R1-2007151, 3GPP, Aug. 2020.
- [23] K. Hogan and R. Jones. (2024). *A Brief Look at Tuning TCP Retransmission Behaviour*. Accessed: Jan. 8, 2026. [Online]. Available: https://services.google.com/fh/files/misc/a_brief_look_at_tuning_tcp_retransmission_behaviour.pdf
- [24] E. Blanton, D. V. Paxson, and M. Allman, *TCP Congestion Control*, document RFC 5681, Sep. 2009. [Online]. Available: <https://www.rfc-editor.org/info/rfc5681>
- [25] F. Abinader, C. Rom, K. Pedersen, S. Hailu, and N. Kolehmainen, "System-level analysis of mmWave 5G systems with different multi-panel antenna device models," in *Proc. IEEE 93rd Veh. Technol. Conf. (VTC-Spring)*, Apr. 2021, pp. 1–6.
- [26] D. Atik, M. Gursu, F. Mehmeti, and W. Kellerer, "Rural handover parameter tuning to achieve end to end latency requirements of future railway mobile communication systems," 2024, *arXiv:2408.06957*.
- [27] R. Temelli, B. Khodapanah, D. Atik, A. Prado, and W. Kellerer, "Supervised learning-based parameter configuration for cell-pair-specific mobility robustness optimization," in *Proc. IEEE 36th Int. Symp. Pers., Indoor Mobile Radio Commun. (PIMRC)*, Sep. 2025, pp. 1–6.
- [28] U. Bauknecht and T. Enderle, "An investigation on core network latency," in *Proc. 30th Int. Telecommun. Netw. Appl. Conf. (ITNAC)*, Nov. 2020, pp. 1–6.
- [29] S. B. Iqbal, A. Awada, U. Karabulut, I. Viering, P. Schulz, and G. P. Fettweis, "Analysis and performance evaluation of mobility for multi-panel user equipment in 5G networks," in *Proc. IEEE 95th Veh. Technol. Conf. (VTC-Spring)*, Jun. 2022, pp. 1–7.
- [30] *Study on Scenarios and Requirements for Next Generation Access Technologies*, Standard TR 38.913, v18.0.0, Apr. 2024.
- [31] *Study on Channel Model for Frequencies From 0.5 to 100 GHz*, Standard TR 38.901, v17.1.0, Jan. 2024.
- [32] A. Gündogan, A. Badalioglu, P. Spapis, and A. Awada, "On the modelling and performance analysis of lower layer mobility in 5G-advanced," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Mar. 2023, pp. 1–6.



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